

# Drought-Induced Deforestation in the Brazilian Amazon<sup>\*</sup>

Tengda Gong<sup>†</sup> Nilesch Shinde<sup>‡</sup> Alankrita Goswami<sup>§</sup> Mahmoud Osman<sup>¶</sup>

AUGUST 2026

---

Drought in the Amazon has become more frequent and intense, contributing to more forest losses in recent years. This paper examines drought-induced deforestation in the Brazilian Legal Amazon and investigates the economics behind it. We focus on agricultural drought to study differential roles of family versus commercial farms in this region. Using high-resolution root-zone soil moisture and land cover data, we document that increases in soil moisture deficits lead to higher deforestation within the same year, with no detectable within-year feedback from local deforestation to local moisture inflow or rainfall once we condition on prior forest stock. Our heterogeneity analyses by family-farm area shares across 770 municipalities reveal the critical role of resource-restricted family farms: (1) drought only raises deforestation where family farming predominates, and not where commercial agriculture dominates; (2) forest-related environmental fines rise rather than fall in high-family-farm municipalities during droughts, ruling out weaker monitoring; (3) high-family-farm municipalities see staple output fall, credit contract, and irrigation fail to expand, while low-family-farm municipalities suffer large sugarcane and cattle losses. Our estimates show that agricultural drought accounts for roughly 257 square kilometers of primary forest clearing during 2006-2019, committing carbon damages of about \$105 million per year.

**JEL Classification:** Q23, Q54, Q12, O13

---

<sup>\*</sup>Corresponding Author: [tengda.gong@umanitoba.ca](mailto:tengda.gong@umanitoba.ca). The authors would like to thank seminar participants at Miami University, Ohio for their valuable feedback.

<sup>†</sup>University of Manitoba. [tengda.gong@umanitoba.ca](mailto:tengda.gong@umanitoba.ca)

<sup>‡</sup>Indiana University Bloomington. [nnshinde@iu.edu](mailto:nnshinde@iu.edu)

<sup>§</sup>University of Manitoba. [alankrita.goswami@umanitoba.ca](mailto:alankrita.goswami@umanitoba.ca)

<sup>¶</sup>NASA Goddard Space Flight Center & ESSIC, University of Maryland, College Park. [mosman1@umd.edu](mailto:mosman1@umd.edu)

# 1 — Introduction

How do climate shocks reshape land-use decisions at tropical forest frontiers? The question lies at the intersection of the economics of climate change, agricultural development, and environmental conservation because the world’s remaining tropical forests are simultaneously major carbon sinks and active margins of agricultural expansion. The Brazilian Legal Amazon exemplifies this tension. It stores one of the largest stocks of terrestrial carbon on Earth while supporting a rapidly evolving agricultural frontier whose land-use decisions are shaped by commodity markets, environmental regulation, and rural development policies (Hanusch [2023](#)). Although Brazil has substantially reduced deforestation through satellite monitoring, law enforcement, supply-chain agreements, and restrictions on rural credit (Gibbs et al. [2016](#); Souza-Rodrigues [2019](#); Heilmayr et al. [2020](#); Baragwanath and Bayi [2020](#); Assunção et al. [2020](#); Assunção, Gandour, and Rocha [2023](#); Baragwanath and Shinde [2025](#)), climate variability has emerged as an increasingly important and comparatively understudied determinant of forest loss (Araujo et al. [2023](#); Araujo, Costa, and Sant’Anna [2025](#); Hsiao et al. [2026](#)).

Drought is particularly important because it is becoming both more frequent and more severe across the Amazon basin. Recent ecological evidence documents increasingly intense soil-moisture deficits and more frequent megadroughts under continued climate change (Anderson et al. [2018](#); Chen et al. [2023](#)). Existing research has primarily examined either the ecological consequences of drought or the reverse relationship through which deforestation alters regional rainfall and moisture recycling (Staal et al. [2020](#); Smith, Baker, and Spracklen [2023](#); Papastefanou et al. [2024](#)). Much less is known about how agricultural drought affects the economic incentives of producers operating at the forest frontier, and in particular whether forest clearing itself constitutes a behavioral response to climate shocks rather than solely a driver of them.

Investigating this question requires understanding how producers respond to drought under different economic environments. In the standard land-use model, forest is cleared whenever the expected return from agricultural production exceeds the value of leaving land under forest (Chomitz and Gray [1996](#); De Janvry, Fafchamps, and Sadoulet [1991](#); Pfaff [1999](#); Rudel et al. [2005](#); Amacher, Ollikainen, and Koskela [2009](#)). A drought lowers agricultural

productivity and expected farm profits, increasing the opportunity cost of converting forest into cropland or pasture. Under this logic, drought should reduce deforestation rather than increase it. Consistent with this prediction, a large literature shows that changes in agricultural productivity alter land-use decisions by changing the profitability of cultivation (Abman and Carney 2020; Carreira, Costa, and Pessoa 2024; Szerman et al. 2024).

Yet a growing empirical literature documents the opposite relationship. Repeated rainfall deficits increase cropland expansion across developing countries (Zaveri, Russ, and Damania 2020; You et al. 2025), while extreme heat induces persistent forest loss at agricultural frontiers by reducing yields and raising agricultural prices (Hsiao et al. 2026; Khan et al. 2025). Similar evidence emerges from Madagascar, West Africa, and Zambia (Desbureaux and Damania 2018; Leblois 2021; He and Chen 2022; Ordóñez et al. 2025; Tabe-Ojong, Tolani, and Molua 2026). These findings overturn the sign of the textbook prediction but not necessarily the underlying logic: adverse weather raises deforestation because reduced yields, given the inelastic demand for subsistence crops, raise crop prices enough to offset the revenue loss, preserving or increasing the expected profitability of agricultural expansion (Hsiao et al. 2026).

The economics of drought-induced deforestation in the Brazilian Amazon, however, may be more complex: producers in this context differ substantially in their ability to absorb drought shocks. Family and commercial farms operate under markedly different production technologies and financial environments. Commercial producers generally have greater access to formal credit, insurance, irrigation, and diversified production systems, allowing them to smooth temporary income fluctuations. Family farms, by contrast, depend primarily on rain-fed production, have limited liquidity, and face substantial borrowing constraints. For these producers, standing forest is not only a productive asset but also a liquid asset that can be converted into cash through timber extraction, pasture conversion, and land appreciation (Rosenzweig and Wolpin 1993; Fafchamps, Udry, and Czukas 1998; Pfaff 1999; Pattanayak and Sills 2001; Amacher, Koskela, and Ollikainen 2009; Jayachandran 2013; Wunder et al. 2014; Ferraro and Simorangkir 2020). When drought reduces current income and informal financial markets fail to insure against the shock, clearing forest provides an immediate source of liquidity. Drought may therefore increase deforestation not because agriculture becomes more profitable, but because the private value of liquidating forest temporarily rises for financially

constrained households.

This distinction generates a sharp empirical prediction. If drought affects deforestation primarily through agricultural profitability, producers suffering the largest production losses should respond similarly regardless of their financial capacity. If, instead, drought operates through a liquidity channel, the response should be concentrated among financially vulnerable producers even when commercially oriented producers experience comparable or larger agricultural losses. The Brazilian Legal Amazon provides an unusually suitable setting to distinguish between these competing mechanisms because family and commercial farms coexist within the same institutional and climatic environment while differing systematically in their ability to cope with drought shocks.

That distinction matters for policy design because the two mechanisms imply fundamentally different interventions. Brazilian conservation policy has been built largely around changing the profitability of forest clearing through satellite monitoring, environmental enforcement, supply-chain agreements, restrictions on rural credit, and property-rights institutions (Cisneros, Zhou, and Börner 2015; Gibbs et al. 2016; Assunção et al. 2020; Baragwanath and Bayi 2020; Heilmayr et al. 2020; Assunção and Rocha 2019; Souza-Rodrigues 2019; Assunção, Gandour, and Rocha 2023; Baragwanath and Shinde 2026). Each of these policies either raises the private cost of clearing or lowers its expected return. If households instead clear forest because drought temporarily tightens liquidity constraints, then rural credit, crop insurance, and drought-indexed transfers become instruments of climate policy, while enforcement alone may leave the underlying incentive unchanged. Distinguishing between profitability and liquidity is therefore central not only to understanding why drought affects deforestation but also to identifying which policy instruments are likely to remain effective as drought shocks become more frequent in the Amazon.

We test these competing explanations using a municipality-by-grid panel covering the Brazilian Legal Amazon from 2005 through 2019. We combine annual forest loss from the 30-meter MapBiomas Collection 9 with a satellite-based measure of agricultural drought derived from daily root-zone soil moisture (RZSM) simulated by NASA’s Global Land Data Assimilation System (GLDAS-2), following Osman et al. (2024).<sup>1</sup> We merge these data with

1. GLDAS-2 is a NASA and NOAA project that creates high-resolution global land surface data. It uses

IBAMA environmental enforcement records, municipality-level agricultural and livestock production, rural credit records, irrigation maps, Agricultural Census data, protected areas, commodity prices, and recently developed atmospheric moisture-flow data. The resulting dataset contains 13,160 municipality-by-grid cells spanning 771 municipalities, yielding approximately 197,400 cell-year observations.

Our identification strategy exploits a key feature of the RZSM data. First of all, agricultural drought is fundamentally a shortage of plant-available water rather than simply low rainfall, making root-zone soil moisture the economically relevant state variable for agricultural production (Wilhite and Glantz 1985). GLDAS-2 simulates soil moisture using meteorological forcing (temperature, precipitation, wind, radiation, etc.) and fixed land-surface characteristics, without incorporating contemporaneous deforestation, irrigation, or groundwater extraction. Consequently, our drought measure is likely to be orthogonal by construction to producers' land-use decisions and adaptation responses. Concerning the small feedback loop between deforestation and rainfall via moisture recycling (e.g., Staal et al. 2020), we find no statistically-detectable within-year feedback from local deforestation to local moisture inflow or rainfall once we condition on prior forest stock in our panel regression model, suggesting that reverse causality from deforestation to drought is unlikely an empirical issue.

Our findings consistently support the liquidity-constraint mechanism. Drought increases forest clearing, but the response is concentrated entirely among municipalities where family farming predominates or family farm area shares are above the median at the baseline. The municipalities with higher concentration of commercial farms exhibit no detectable increase in deforestation despite experiencing the same drought shocks. Moreover, the response operates almost exclusively along the intensive margin, increasing the amount of clearing where deforestation is already occurring rather than substantially expanding clearing onto previously untouched land. These patterns suggest that drought does not broadly increase the profitability of agricultural expansion. Instead, it induces additional clearing precisely among producers most likely to face binding liquidity constraints.

satellite and ground measurements with land surface models to calculate water and energy states on land. Osman et al. (2024) use its RZSM data to measure flash drought.

The mechanism evidence reinforces this interpretation. Agricultural losses occur across both production systems but translate into forest clearing only among financially vulnerable producers. Rice, the principal rain-fed staple cultivated by family farms, experiences large reductions in planted area, harvested area, production, and revenue during droughts in high-family-farm municipalities where drought-induced deforestation occurs. Moreover, drought does not increase rice prices in these municipalities but in low-family-farm municipalities, further ruling out the profitability mechanism. Commercial producers also experience substantial losses—most notably in sugarcane production and cattle holdings—yet do not increase forest clearing. The deforestation response therefore tracks financial vulnerability rather than the magnitude of agricultural losses, distinguishing the liquidity channel from a pure profitability mechanism.

The remaining coping margins tell a consistent story. Rather than expanding adaptation investments during droughts, high-family-farm municipalities where clearing rises experience contractions in rural credit, including PRONAF credit, Brazil’s principal rural credit program for family farms, while irrigation does not expand. In high-family-farm municipalities producing both rice and cassava (staple crops), producers tend to harvest more drought-tolerant cassava in response to early-season drought that reduces rice planted area, suggesting that family farms may exhaust relatively inexpensive production adjustments before turning to forest clearing as a source of liquidity.<sup>2</sup> Together, these results indicate that the institutions capable of smoothing temporary income shocks weaken precisely where drought-induced deforestation is most pronounced.

Several additional findings rule out leading alternative explanations. Environmental enforcement intensifies rather than weakens during droughts: both the number and value of IBAMA forest-related fines increase, particularly in municipalities where deforestation rises, rejecting the hypothesis that the estimated effects reflect reduced monitoring. Likewise, the timing of the response is inconsistent with a flammability-based explanation. Pre-growing-season drought either reduces or has no effect on clearing, whereas deforestation rises only in response to growing-season moisture deficits that adversely affect the actual agricultural production in the field.

2. Cassava is a multi-year crop with flexible harvesting time.

Our paper makes three contributions to the literature. First, we provide causal evidence on drought-induced deforestation in the Brazilian Amazon, to the best of our knowledge. Staal et al. (2020) documents a positive association between annual deforestation and cumulative water deficits using within-cell variation over time as we do. We additionally control for important confounders, including municipality-specific exposures to commodity prices, policies, protected areas, and prior forest stock. Pimenta, Reimer, and Novan (2025) also studies drought-induced deforestation with a focus on public versus private land tenure, although they do not control for most of the listed confounders.<sup>3</sup> We focus on family versus commercial farms and find family farms tend to clear forest in response to drought while commercial farms may not even in municipalities with high area shares of public forest.

Second, we provide evidence that distinguishes the liquidity channel from the profitability channel for the drought-induced deforestation in the Brazilian Amazon. While recent work shows that adverse weather can accelerate forest clearing (Hultgren et al. 2025; Tabe-Ojong, Tolani, and Molua 2026; Khan et al. 2025; Hsiao et al. 2026), existing explanations primarily emphasize on inelastic food demand for subsistent crops. We show instead that drought-induced deforestation is concentrated among financially constrained producers, even when commercially oriented producers experience comparable or larger production losses. Jayachandran (2013) and Ferraro and Simorangkir (2020) also find evidence on the liquidity channel for deforestation in Uganda and Indonesia where smallholders dominate.

Third, we contribute to the growing literature on climate adaptation and conservation policy. Our results imply that policies designed to discourage profitable forest conversion; including monitoring, enforcement, and market-based restrictions; may not fully address deforestation arising from temporary liquidity shocks (Assunção et al. 2020; Assunção et al. 2023; Araujo, Costa, and Sant’Anna 2025). Because deforestation may reflect households’ efforts to smooth income following drought, expanding access to rural finance, insurance, or drought-indexed transfers can complement traditional conservation policies by improving the resilience of vulnerable agricultural households. Our estimates show that agricultural drought accounts for roughly \$105 million in carbon damages per year, suggesting substantial

3. They assume that public and private land share same shocks within each cell. Their 0.5-degree cells, however, can be larger than a municipality in the Brazilian Amazon.

room for such policy interventions.

The remainder of the paper proceeds as follows. Section 2 describes the institutional and environmental setting. Section 3 develops the conceptual framework. Sections 4 and 5 describe the empirical strategy and data. Sections 6 and 7 present the main results and mechanism evidence. Section 8 reports robustness analyses, Section 9 discusses welfare and policy implications, and Section 10 concludes.

## 2 — Background

**Deforestation in the Brazilian Amazon.** The Brazilian Legal Amazon spans approximately five million square kilometers across nine states and contains the largest contiguous block of tropical forest on Earth. Between 2000 and 2019 the region lost more than 400,000 square kilometers of forest cover, making it one of the most active deforestation frontiers globally and a major source of land-use carbon emissions (Souza Jr. et al. 2020; MapBiomass Project 2024; Ometto et al. 2023). Deforestation rates are highly volatile over time. After peaking in the early 2000s at over 25,000 km<sup>2</sup> per year, annual forest loss fell sharply following the introduction of the Action Plan for the Prevention and Control of Deforestation in the Legal Amazon (*PPCDAm*) in 2004 and remained comparatively low through the early 2010s, before rising again toward the end of the decade (West and Fearnside 2021). This temporal variation reflects the combined influence of commodity prices, environmental governance, land tenure conditions, and infrastructure expansion (Assunção, Gandour, and Rocha 2015; Souza-Rodrigues 2019; Assunção et al. 2020; Baragwanath and Bayi 2020; Baragwanath and Shinde 2025). Understanding what drives deforestation in any given year thus requires isolating the contribution of each channel—a challenge that motivates the empirical design of this paper.

**Deforestation policy.** Brazil has deployed a layered set of policy instruments to reduce forest loss in the Amazon. The 2006 Amazon Soy Moratorium, a voluntary supply-chain agreement, committed major traders not to purchase soy grown on land cleared after that date, effectively extending zero-deforestation requirements to a large share of the commercial



soy sector (Gibbs et al. 2016; Heilmayr et al. 2020; Assunção, Gandour, and Rocha 2015; Assunção et al. 2020). The Priority Municipalities (*Municípios Prioritários*, or MPs) program, introduced in 2008, placed the highest-deforesting municipalities on a public blacklist, triggering enhanced monitoring, credit restrictions, and intensified enforcement by IBAMA, the federal environmental agency (Cisneros, Zhou, and Börner 2015; Assunção and Rocha 2019; Baragwanath and Shinde 2025). The 2012 Forest Code established farm-level requirements for maintaining Legal Reserves of native vegetation and Permanent Protection Areas, while also offering amnesty for historical deforestation. Together these programs substantially altered the incentive landscape for land use in the Amazon, and their implementation dates provide important sources of variation that we control for throughout the analysis. We also account for a major 2008 rural credit concession that restricted subsidized credit access for producers within the Amazon Biome, a policy shock that coincided with the Priority Municipalities program and confounds simple before-after comparisons (Assunção et al. 2020; Souza, Herschmann, and Assunção 2020).

**Agricultural structure: family farms and commercial producers.** The agricultural economy of the Legal Amazon is characterized by two fundamentally different production systems that face distinct financial and technological constraints. At one end are large commercial operations producing soybeans, cattle, and other export-oriented commodities. These farms account for most agricultural land and are the primary targets of environmental monitoring, supply-chain commitments, and formal credit programs. At the other end are family farms, which dominate numerically across much of the Amazon frontier and specialize in labor-intensive production of staple food crops for domestic markets.<sup>4</sup>

These differences imply fundamentally different margins of adjustment following adverse climate shocks. Commercial producers generally have access to formal credit, crop insurance,

4. Under Brazil’s Family Farming Law (Lei 11.326/2006), family farms are defined as establishments that rely primarily on household labor, derive most household income from the farm, and operate below a municipality-specific size threshold of four fiscal modules. According to the 2006 Agricultural Census, family farms accounted for approximately 84 percent of all agricultural establishments in Brazil, although they occupied only about one-quarter of agricultural land. They produce a disproportionate share of staple crops, including cassava, beans, maize, and rice, which are central to Brazil’s domestic food supply. Please refer, Assunção et al. (2020), Souza, Herschmann, and Assunção (2020), Leal, Assunção, and Hoover (2026), Souza and Albuquerque (2023), and Instituto Brasileiro de Geografia e Estatística (IBGE) (2006)

irrigation, and diversified production portfolios that allow temporary income losses to be absorbed without substantial changes in land use. Family farms, by contrast, depend predominantly on rain-fed agriculture, possess limited liquid assets, and face persistent borrowing constraints.<sup>5</sup> These differences motivate the central hypothesis of this paper. When drought reduces agricultural income, commercially oriented producers can smooth consumption through financial markets, whereas liquidity-constrained family farms may instead rely on assets held on the farm itself. In frontier environments, standing forest constitutes one such asset: it can be converted into immediate income through timber extraction, pasture expansion, or by increasing the marketability of land claims. Consequently, identical climate shocks may generate very different deforestation responses depending not on the magnitude of production losses, but on producers' ability to finance those losses without liquidating forest.

**Drought in the Amazon.** The Legal Amazon has experienced a marked increase in drought frequency and severity over the past two decades. Three major droughts — in 2005, 2010, and 2015–2016 — caused widespread ecological disruption, elevated fire incidence, and large-scale tree mortality across the basin (Anderson et al. 2018; Otkin et al. 2018; Aragão et al. 2018; Yuan et al. 2023). These events were associated with anomalous sea surface temperatures in the tropical Atlantic and Pacific and are consistent with projected trends toward a drier and more variable Amazon climate under continued greenhouse gas forcing (Otkin et al. 2018; Osman et al. 2024; Yuan et al. 2023). Root zone soil moisture deficits during these events were severe and prolonged, with cumulative annual deficits well above historical norms across large portions of the basin. While the ecological consequences of Amazon drought are extensively documented, the behavioral and economic responses of agricultural producers to drought-induced income stress have received comparatively little attention. This paper focuses on that gap, asking whether and how soil moisture deficits translate into changes in land-use decisions at the farm level.

5. The 2006 Agricultural Census reports substantial disparities in access to productive services between family and commercial farms. Fewer than one-fifth of family establishments reported receiving technical assistance, compared with more than 60 percent of non-family farms. Access to formal rural credit and irrigation infrastructure is similarly concentrated among larger commercial producers (Instituto Brasileiro de Geografia e Estatística (IBGE) 2006)

**Drought, income shocks, and land use.** In rain-fed agricultural systems, soil moisture deficits reduce crop yields and depress farm revenues, generating income shocks that are difficult to insure against in markets characterized by limited financial development. A substantial literature in development economics documents that households and firms facing negative income shocks adopt coping strategies that prioritize short-term liquidity over long-run efficiency, including asset sales, labor reallocation, and changes in production choices (Chomitz and Gray 1996; De Janvry, Fafchamps, and Sadoulet 1991; Pfaff 1999; Rudel et al. 2005; Amacher, Ollikainen, and Koskela 2009). In agricultural frontier regions, forests represent a particularly flexible asset in this context: clearing can generate immediate returns through timber sales, pasture conversion, or speculative land appreciation, and the returns can be realized quickly relative to the time horizon of formal credit or insurance instruments.

This coping logic implies that drought may increase deforestation not by making agriculture more profitable in expectation, but by temporarily raising the relative attractiveness of timber and land conversion income for producers facing short-term cash constraints. If correct, this mechanism has an important implication for conservation policy: enforcement and market-based tools designed for stable economic environments may be insufficient to prevent deforestation during periods of acute climate stress, when the private returns to forest clearing are elevated precisely because normal agricultural income channels are disrupted. The empirical analysis that follows tests this mechanism using variation in drought severity across Legal Amazon municipalities and exploits the sharp structural divide between family farms and commercial producers to identify the economic channel.

### 3 — Conceptual Framework

This section formalizes the coping-response mechanism and derives the empirical predictions that guide the analysis. Our goal is not a fully specified structural model but a transparent framework that clarifies why drought can raise deforestation, and why the response should be concentrated among financially constrained producers.

**Set-up.** Consider a frontier producer  $i$  who at the start of period  $t$  holds a stock of standing forest and chooses how much of it to clear,  $h_{it} \geq 0$ .<sup>6</sup> Clearing yields a net monetary payoff  $R(h_{it})$  from timber sales, pasture conversion, and land appreciation, with  $R' > 0$  and  $R'' < 0$ . Standing forest also carries a continuation value  $V$  (future agricultural rents, ecosystem services, and the option value of delaying an irreversible decision), so clearing an extra unit of forest forgoes  $V$ . Absent frictions, the producer clears up to the point where  $R'(h_{it}) = V$ , and a negative agricultural shock that lowers expected farm returns tends to lower  $R'(h_{it})$  relative to  $V$ , *reducing* clearing. This is the standard opportunity-cost logic, and it predicts that drought should slow deforestation. We acknowledge that farm output prices, especially for staple crops, may rise sharply in response to drought that reduces output supply, so that  $R'(h_{it})$  can even increase instead (Hsiao et al. 2026).

**Liquidity constraint.** The key departure is that the producer must also finance current consumption and production out of period- $t$  resources. Let  $y_{it}$  denote realized farm income and suppose the household has a subsistence or working-capital requirement  $\bar{c}$  that must be met each period, with limited access to credit or insurance to bridge shortfalls. When income is adequate ( $y_{it} \geq \bar{c}$ ) the constraint does not bind and the producer follows the frictionless rule describe above. When a drought drives income below the threshold ( $y_{it} < \bar{c}$ ), the constraint binds, and the shadow value of immediate cash from timber sales per unit of cleared forest,  $L'(h_{i,t}|y_{i,t}) > 0$ , enters the clearing decision:  $R'(h_{it}) + L'(h_{i,t}|y_{i,t}) = V$ . A binding liquidity constraint scales up the marginal value of the immediate cash that clearing generates. Drought may therefore operate through two offsetting channels: it lowers  $R'(h_{it})$ , which discourages clearing; but for liquidity-constrained producers ( $y_{it} < \bar{c}$ ), it also raises  $L'(h_{i,t}|y_{i,t})$ , which encourages clearing. Which channel dominates is an empirical question, and its answer differs sharply across producer types, as illustrated below.

**Heterogeneity and predictions.** Commercial producers have access to formal credit, crop insurance, irrigation, and diversified production portfolios; for them  $L'(h_{i,t}|y_{i,t}) \approx 0$  even in drought years, so the opportunity-cost channel dominates and drought does not necessarily

6. Depending on the location, she or he may or may not face a margin of undesignated public forest, claimed by the state but not allocated, which is in practice open access.

raise clearing. Family farms are rain-fed, uninsured, and credit-constrained; for them drought may push  $y_{it}$  below  $\bar{c}$  and thereby raise  $L'(h_{i,t}|y_{i,t})$ , i.e., the liquidity channel kicks in and will dominate when drought is sufficiently severe, such as megadrought. This yields the framework’s central prediction and three auxiliary ones that we take to the data:

- (P1) **Concentration.** The deforestation response to drought is positive and concentrated among family-farm dominated municipalities, and absent among commercial-farm dominated municipalities.
- (P2) **Vulnerability, not loss size.** The response tracks financial vulnerability rather than the magnitude of production loss: commercial farms can suffer large drought losses without clearing, as long as it can smooth the shock.
- (P3) **Coping margins.** Where the response is largest, the margins that would otherwise absorb the shock—agricultural output, rural credit, irrigation—are impaired rather than expanded.
- (P4) **Real clearing, not measurement.** Because clearing is a behavioral response and is measured from satellite imagery, drought-induced deforestation should be accompanied by more enforcement activity, ruling out a monitoring-failure explanation.

The remainder of the paper tests P1–P4 using data. The sharp structural divide between family and commercial farms in the Brazilian Amazon, as illustrated in Section 2, provides the suitable context for these tests. Before jumping into the data section, we outline our empirical framework in the next section.

## 4 — Empirical Framework

We use the following dynamic panel regression model to estimate the effects of agricultural drought at the municipality-RZSM (root zone soil moisture) grid or municipality level, depending on the outcomes of interest:

$$Y_{j,t} = \alpha Y_{j,t-1} + \beta DI_{j,t} + C'_{j,t}\gamma + FE_j + FE_t + \varepsilon_{j,t},$$

where (1)  $Y$  denotes outcomes of interest, including deforestation (municipality-RZSM grid level), deforestation related environmental fines (municipality level), crop and cattle production (municipality level), irrigation and rural credit use (municipality level); (2)  $DI$  denotes the RZSM-based drought intensity (see details in the data section); (3) the control vector  $C$  includes municipality-RZSM grid-specific or municipality-specific prior forest stock, protected land area shares, municipality-specific exposure to exogenous crop and cattle prices as well as major municipality or regional level deforestation-related regulation policies (see details in the data section); and (4)  $FE$  represents individual (municipality-RZSM grid or municipality) and year fixed effects.

As illustrated below, our primary drought intensity is derived from RZSM data at a resolution of 0.25 degree, corresponding to grid cells of approximately 28 km at the equator. Hence, a municipality can be larger than a RZSM grid cell. Given that, we use the area-weighted drought intensity in the regression when it comes to municipality-level outcomes. For municipality-RZSM grid-level outcomes, we use our primary drought intensity directly.

In our main specification, we cluster standard errors at the municipality level, allowing for arbitrary serial correlation within each municipality over time. To account for unobserved or unmeasured spatially correlated shocks, we use Conley spatial standard errors with distance cutoffs ranging from 200 to 500 km as a robustness check. Our drought-deforestation results remain robust. See detailed results in the Appendix.

We include the lagged outcome variable to account for the strong persistence in our outcome variables. We choose a dynamic panel regression as a static panel regression will otherwise produce an estimand with a bias that is notably larger than the dynamic panel regression when the outcome variable is highly autocorrelated (Klosin 2024). Moreover, because our long panel spans fifteen years, the traditional Nickell bias associated with a dynamic panel regression tends to be small (Klosin 2024).

Staal et al. (2020) provide model-based evidence of a small feedback from deforestation to drought in the Amazon. Using a moisture-tracking algorithm, they show that deforestation reduces forest evapotranspiration and thereby lowers downwind rainfall, increasing dry-season water deficits during 2003-2014.<sup>7</sup> Smith, Baker, and Spracklen (2023) show that cumulative

7. Staal et al. (2020) also documents a positive association between annual deforestation and cumulative

tropical deforestation over a long period reduces precipitation in neighboring grids, with stronger effects at larger spatial scales. Using cumulative forest loss from 2003 to 2017, they find robust rainfall reductions at scales greater than 50 km, with the largest estimated reduction at 200 km. Our deforestation measure is at the  $0.25^\circ$  resolution, corresponding to roughly 28 km at the equator. Also, annual deforestation is small relative to long-term cumulative deforestation. Therefore, the feedback from deforestation to drought is less of a concern in our setting. As expected, we find a negligible and statistically insignificant association between our annual deforestation measure and local moisture inflow, obtained from a recent global moisture-flow dataset (Zhang et al. 2025), and local rainfall, obtained from TerraClimate. See detailed results in the Appendix.

## 5 — Data

We combine datasets of land cover, root-zone soil moisture (RZSM), irrigation map, administrative environmental fines and agricultural statistics as well as rural credit use to construct two panels for the Brazilian Legal Amazon: one at the municipality-RZSM grid level and one at the municipality level. We use the first panel to document the drought-induced deforestation at a high spatial resolution. We use the second panel to examine economic mechanisms behind it, with a focus on family versus commercial farms.

We analyze the time period from 2005 to 2019. The starting year places our sample entirely within the policy regime established by the Action Plan for the Prevention and Control of Deforestation in the Legal Amazon (PPCDAm), launched in 2004 (West and Fearnside 2021). PPCDAm reorganized federal monitoring, enforcement, and land tenure policy in the Amazon, and the blacklisting and credit-restriction programs we control for below were introduced as instruments under it. Beginning in 2005 therefore holds the broad enforcement architecture fixed, so that our estimates capture responses to drought within a stable policy environment rather than a mix of drought effects and the transition into a water deficits using within-cell variation over time. We aim to provide causal evidence on the drought-induced deforestation by controlling for municipality-specific exposures to commodity prices, policies, protected areas, and prior forest stock, as discussed above.

new regime.<sup>8</sup> Two data constraints point the same way: municipal agricultural statistics and environmental enforcement records are sparse before 2005, and our rural credit data end in 2019. Additionally, because the inclusion of the lagged outcome in our dynamic panel regression requires an initial year to construct the first lag, our effective estimation sample spans 2006–2019.

## 5.1. Agricultural Drought

Agricultural drought occurs when soil moisture is insufficient to meet the water demands of plants (Wilhite and Glantz 1985). We therefore measure drought using root zone soil moisture (RZSM) rather than a precipitation-based index.

We obtain daily RZSM from NASA’s Global Land Data Assimilation System, version 2 (GLDAS-2), which combines satellite observations, ground-based meteorological measurements, and the Catchment Land Surface Model (CLSM) to produce physically consistent estimates on a 0.25-degree grid, roughly 28 km at the equator. We use the series compiled by Osman et al. (2024), who use it to measure global flash drought.<sup>9</sup> RZSM is reported in kilograms per square meter, the total water mass held in the root zone soil column (0–100 cm below surface). This is the layer directly relevant for crop water uptake and plant stress.

Two features of the CLSM make RZSM well suited to identifying exogenous water shocks. First, the model uses a static land cover classification. Specifically, soil and vegetation types do not respond to deforestation, our main outcome.<sup>10</sup> Second, the model does not represent irrigation or groundwater pumping. Our drought measure therefore reflects hydroclimatic water stress generated by meteorological forcing and fixed land surface characteristics, not

8. Because PPCDAm applied to the entire Legal Amazon at once, it generates no cross-municipality variation and is absorbed by year fixed effects. Restricting the sample to the post-launch period is what makes this absorption meaningful.

9. Osman et al. (2024) use the RZSM field derived from the Catchment Land Surface Model (CLSM) version 2.5. This dataset provides a comprehensive and continuous daily record of RZSM, offering an extensive temporal scope for analyzing environmental and climatic soil moisture trends from 1980 to 2021. They select GLDAS-2 for its enhanced hydrological representation, incorporating GRACE data assimilation to uniquely capture the complex interactions between soil moisture and groundwater for a more comprehensive understanding of the water balance. We use historical RZSM distributions from this dataset to estimate grid-specific drought thresholds based on the bottom 10th percentiles.

10. Base soil and vegetation types are fixed throughout the sample period, although other land surface model parameters may capture historically repeated seasonal cycles within a year.



human adaptive responses.

We construct drought intensity in three steps. We standardize daily RZSM within each grid cell using its full historical record (from 1980 to 2021). We then compute daily deficits relative to the cell-specific 10th percentile. We finally sum daily deficits at the grid-cell level within a year or a crop season (details below). Larger values indicate more severe or more prolonged soil moisture stress.

We use drought intensity in two panels: one at the municipality-RZSM grid level, one at the municipality level. For the latter, we use area weights to aggregate raw RZSM-grid level drought intensities within each municipality. The mean annual deficit at the municipality-RZSM grid level is about 3.0 units, but the distribution has a long right tail reaching 548 units in the most extreme cell-years, and 15 percent of cell-years fall below the drought threshold on at least one day.<sup>11</sup> Municipality-level annual RZSM deficits have similar statistics with a smaller variance, as reported in Table A1.

We use annual RZSM deficits in our main drought-deforestation and drought-environmental fines regressions. When it comes to the production effects for major temporary crops, such as rice and soybean, we separate drought by crop season to study agricultural mechanisms. Following Rattis et al. (2021), we define a general growing season in the Legal Amazon from September of  $t - 1$  to June of year  $t$ , and a general pregrowing season from July of year  $t - 1$  to August of year  $t - 1$ .<sup>12</sup> Growing season deficits affect crop development from germination through maturation, while pre-growing season deficits precede the planting window and may shape planting decisions. We also use these seasons to explore certain adaptation margins, such as crop substitution, irrigation and rural credit use.

### *Why Not SPEI?*

The Standardized Precipitation Evapotranspiration Index (SPEI) is the most widely used drought index in economics (Vicente-Serrano, Beguería, and López-Moreno 2010; Reig-Gracia and Latorre Garcés 2023). We prefer RZSM for two reasons. First, RZSM measures a stock

11. This drought percentage is higher than 10%, as the weather gets hotter and drier in more recent years.

12. USDA uses the same growing season for soybean, mostly grown in the Legal Amazon. Assunção, Gandour, and Rocha (2015) use a similar-but-slightly-shorter growing season, from August of year  $t - 1$  to May of year  $t$ .

of plant-available water, whereas SPEI measures a flow of atmospheric water balance. Crops draw on accumulated soil moisture, not on contemporaneous rainfall alone. Second, GLDAS computes actual evapotranspiration from the modeled water balance, whereas SPEI uses potential evapotranspiration. Actual evapotranspiration is closer to the water stress vegetation experiences.<sup>13</sup> Nevertheless, in Section 8, we show our drought-induced deforestation results hold when using drought deficits derived from SPEI at 0.25-degree resolution.

### *Deforestation and Drought*

Deforestation can in principle affect rainfall through moisture recycling, which would complicate a causal interpretation (Staal et al. 2020; Smith, Baker, and Spracklen 2023). The feedback documented in that literature operates over large spatial domains and long time scales, whereas we study year-over-year variation at 0.25 degrees, and annual clearing is small relative to cumulative forest loss. Consistent with this, Section 8 shows that annual local forest loss has a limited and statistically insignificant association with local moisture inflow, taken from the global moisture source-sink dataset of Zhang et al. (2025), and with local rainfall from TerraClimate (Abatzoglou et al. 2018), once we condition on prior forest stock. Prior forest stock itself is strongly associated with both, which is the reason we control for it throughout. We also report Conley spatial standard errors with 200 and 500 kilometer cutoffs.

## **5.2. Deforestation**

Our primary outcome is annual forest loss measured against a fixed forest baseline, constructed from MapBiomas Collection 9 annual land use and land cover data at 30-meter resolution (Souza Jr. et al. 2020; MapBiomas Project 2024). MapBiomas combines Landsat imagery, machine learning classification, and expert validation, and covers Brazil from 1985 onward.

13. For example, during rapid-onset or prolonged droughts, atmospheric demand (PET) increases sharply, leading to extremely low SPEI as the index assumes maximum atmospheric water loss. In reality, however, stressed plants may close their stomata to conserve their water supply, which can significantly reduce actual evapotranspiration (AET). Because plants transpire less, they withdraw less water from the root zone—a biological self-preservation mechanism that slows the rate of RZSM depletion. GLDAS-modeled RZSM explicitly accounts for this constrained AET, making it a more realistic proxy for the actual hydro-agricultural stress experienced by vegetation than SPEI.

We identify forest formation pixels in the base year 2000 and measure annual loss from that fixed stock over 2005–2019, aggregating to the municipality-grid-year level.<sup>14</sup> Measuring loss against a fixed baseline rather than as a year-to-year net change avoids offsetting clearing against subsequent regrowth, so the outcome reflects conversion of the forest standing in 2000 rather than net cover change. As shown in Table A1, mean annual loss is 0.98 km<sup>2</sup> per cell-year, with a maximum of 85 km<sup>2</sup>; 36 percent of cell-years record no loss. We therefore apply the inverse hyperbolic sine transformation, which accommodates zeros while approximating the logarithm at larger values. We also report an extensive-margin specification in which the outcome is an indicator for any loss in a given year.

### 5.3. Environmental Enforcement

Increases in measured deforestation during droughts could reflect changes in clearing behavior or changes in monitoring. To distinguish these, we use administrative records on forest-related environmental fines issued by the Brazilian Institute of Environment and Renewable Natural Resources (IBAMA). The data report the annual number and monetary value of infractions for illegal deforestation, forest degradation, and unauthorized clearing at the municipality level. As shown in Table A1, the mean municipality-year records 7.2 fines worth R\$2.9 million, and roughly half of municipality-years record no fine at all, so both measures enter in inverse hyperbolic sine form. We also report an extensive-margin specification using an indicator for any fine.

These records are administrative rather than remotely sensed and so provide an independent measure of government activity. An increase in fines during drought years is inconsistent with a monitoring-failure interpretation and supports the view that observed increases in clearing reflect genuine changes in behavior.

### 5.4. Who Farms Each Municipality?

A central dimension of heterogeneity in our analysis is the prevalence of family farming. We measure it using the 2006 Brazilian Agricultural Census, the most comprehensive source of

14. Our drought-induced deforestation results are robust to alternative base years of 1985 and 1995.

farm structure available for our period ([Instituto Brasileiro de Geografia e Estatística \(IBGE\) 2006](#)). Brazilian law (Lei 11.326/2006) defines family farms by farm size, reliance on family labor, and household dependence on agricultural income.

For each municipality we calculate the share of agricultural area operated by family farms, which averages 33 percent and ranges from essentially zero to 97 percent across the 770 municipalities present in the Census. We classify municipalities above the Legal Amazon median as family-farm intensive and the remainder as not. This classification is fixed across all years. Because it is measured before our sample period, it captures pre-existing differences in agricultural organization rather than endogenous responses to the droughts we study. Total agricultural establishment area from the same Census also serves as the denominator for our irrigation measures below.

## 5.5. Agricultural Production

We measure crop production using the *Pesquisa Agrícola Municipal* (PAM), an annual survey conducted by the Brazilian Institute of Geography and Statistics (IBGE) ([Instituto Brasileiro de Geografia e Estatística \(IBGE\) 2023a](#)). PAM reports planted area, harvested area, output quantity, and output revenue at the municipality level. We divide output revenue by output quantity to recover average price.

We focus on four temporary crops that span the range of production systems in the Legal Amazon. Soybean and sugarcane are grown predominantly at commercial scale. Rice and cassava are the primary staple crops of smallholder and family farm households.<sup>15</sup> We restrict each crop sample to municipalities reporting positive production throughout the period, which yields 167 municipalities for soybean, 178 for sugarcane, 496 for rice, and 689 for cassava. Production outcomes enter in logs, and each regression is weighted by the 2005 municipality-specific level of the outcome. See summary stats of these outcomes in [Table A2](#).

Drought exposure is defined to match each crop’s production cycle. Soybean and rice are annual crops, so we use separate pre-growing and growing season deficits. Cassava and

15. We exclude first and second corns, another major crops in the Legal Amazon, because growing seasons differ between first and second corns, making a common drought exposure window difficult to define. Moreover, states dramatically change corn portfolios over time. Without production data by corn types, is it empirically infeasible to conduct meaningful regression analyses.

sugarcane are continuous multi-year crops, so we use the year-round deficit. A separate sample of 464 municipalities that jointly cultivate rice and cassava lets us test for substitution toward the more drought-tolerant crop.

We measure livestock using the *Pesquisa Pecuária Municipal* (PPM), also compiled annually by IBGE ([Instituto Brasileiro de Geografia e Estatística \(IBGE\) 2023b](#)). We focus on cattle head counts, the dominant livestock activity in the region, restricting to the 767 municipalities with consistently positive reported stocks. See the summary statistics in Table A3. Head counts enter in logs and regressions are weighted by 2005 head count. Cattle allow us to test whether drought-induced income shocks extend beyond crops, plausibly through forage deterioration.

## 5.6. Rural Credit

Liquidity constrained producers may borrow to smooth income losses ([Jayachandran 2013](#)). Rural credit is the primary policy instrument for financing agricultural production in Brazil. Annual planned rural credit is roughly 30 percent of total agricultural production value, and increases in credit supply raise production, productivity, and native forest cover, particularly in the Amazon Biome ([Assunção et al. 2020](#)).

We obtain credit data from two sources. For 2005–2011 we use the municipality-level series assembled by [Assunção et al. \(2020\)](#), which reports credit by crop and by cattle. For 2014–2019 we use loan-level records from the Central Bank of Brazil’s Rural Credit and Proagro Operations System (SICOR) ([Central Bank of Brazil \(BCB\) 2023](#)), which identify both the purpose and the credit program of each operation. We aggregate loans to the municipality-year level using the agricultural-year definition of [Assunção et al. \(2020\)](#), August of year  $t - 1$  through July of year  $t$ , which closely matches our crop-year window. Because that window requires data from the preceding August, the SICOR series yields complete measures only from 2014, and 2012 and 2013 are unavailable in both sources. Credit regressions therefore omit those years.

We construct total credit amounts and contract counts for crop and cattle production across the full credit sample, and separately identify two programs of particular interest within the SICOR period. PRONAF (*Programa Nacional de Fortalecimento da Agricultura*

*Familiar*) is the principal federal credit program for family farms, which matters because we find drought effects concentrated in family-farm intensive municipalities. Proirriga (*Programa de Financiamento à Agricultura Irrigada e ao Cultivo Protegido*) finances irrigation and protected cultivation, and is used sparsely: mean annual disbursement is roughly R\$61,000 per municipality and the mean contract count is 0.06, as shown in Table A3. All credit variables enter in inverse hyperbolic sine form to accommodate municipality-years with no operations.

## 5.7. Irrigation

We measure irrigation using the MapBiomas Irrigation Collection, which provides annual 30-meter maps of irrigated agriculture across Brazil from 1985 onward. The dataset is derived from Landsat imagery using machine learning classification and expert validation, and each irrigated pixel carries a code linking it to the underlying land use class in the MapBiomas Land Use and Land Cover Collection, so we observe the crop associated with each irrigated pixel.

We aggregate irrigated pixels to the municipality-year level and divide by total agricultural establishment area from the 2006 Census, giving an irrigated area share that is comparable across municipalities of very different size. We construct this share for soybean, rice, sugarcane, all other temporary crops. MapBiomas reports no irrigation for cassava, consistent with cassava being rain-fed throughout the Legal Amazon. As shown in Table A3, irrigation is limited in absolute terms: the mean share across all temporary crops is 0.09 percent of farm area. These measures let us test whether producers adopt irrigation in response to drought and whether irrigation attenuates drought effects on production and clearing.

## 5.8. Prices and Policy Controls

Drought may coincide with changes in agricultural profitability. Following Assunção, Gandour, and Rocha (2015) and Assunção et al. (2020), we control for exogenous commodity price variation. Price series come from the Paraná State Secretariat of Agriculture and cover beef cattle, soybean, cassava, rice, corn, and sugarcane. Because prices are set outside local markets,

they reflect demand-driven variation rather than local supply responses. We weight them by each municipality’s crop composition in a fixed base period of 2004–2005, and construct separate indices for each semester of the agricultural year, entered contemporaneously and with one lag. The crop index summarizes crop-specific prices by their first principal component; the cattle index enters in levels. Regressions for crop-specific production outcomes use the corresponding crop price directly.

We additionally control for the major conservation policies in force during our period, most of which were introduced as instruments of PPCDAm: blacklisting under the Priority Municipalities (*Municípios Prioritários*) program, which covers 3.5 percent of municipality-years; participation in the Green Municipality (*Município Verde*) program, at 8.7 percent; and the 2008 rural credit restriction and soy moratorium within the Amazon Biome, which applies to 52 percent. Unlike PPCDAm itself, these apply to some municipalities and not others and are therefore separately identified from year fixed effects. The first two enter with a one-year lag to avoid simultaneity with current deforestation. We also control for the lagged municipality share of protected land, which averages 23 percent, using the World Database on Protected Areas. Every specification additionally includes the lagged inverse hyperbolic sine of forest stock, which bounds potential future loss and, as Section 8 shows, absorbs most of the association between forest cover and local moisture.

## 5.9. Summary Statistics

Table A1 reports summary statistics for main variables. Tables A2 and A3 reports summary statistics for the mechanism outcomes. Table A4 reports summary statistics for the price and policy controls. We already discuss some of data stats in previous sections. In the following, we focus on Table A1, which summarizes our deforestation, environmental fines, and agricultural drought measures.

Both deforestation and forest-related environmental fines are widely dispersed. Annual clearing is strongly right-skewed: the mean cell-year loses about 1 km<sup>2</sup> of forest, but the standard deviation is three times the mean and the maximum exceeds 80 km<sup>2</sup>. Fine data show the same pattern, with a mean of 7 per municipality-year against a maximum of 841. As mentioned before, both deforestation and fine data have zero values for certain fractions

of observations (36% and 50%). Given these data features, we transform both outcomes using the inverse hyperbolic sine. We also supplement our regression analyses using dummy variables to study the extensive effects of agricultural drought on these two main outcomes.

We directly use the drought intensity in our regressions without transformation, as the underlying RZSM is already standardized. At the municipality-RZSM grid level, more than 15% of cell-years experience some drought. At the municipality level, this number increases to 26%. Both incidence rates are larger than 10%, indicating more frequent drought in recent years. We acknowledge that drought intensity is highly skewed, with the maximum exceeding at least 80 times its mean. We check the sensitivity of our results to the associated megadrought.

## 6 — Main Results

### 6.1. Agricultural Drought and Deforestation

Table B1 reports the average and heterogeneous effects of year-round root zone soil moisture (RZSM) deficits on deforestation across Legal Amazon municipalities from 2005 to 2019. As outlined in Section 4, both specifications include municipality-by-RZSM-grid fixed effects, year fixed effects, the lagged deforestation, the lagged forest stock, municipality-specific exposure to exogenous crop and cattle prices, protected land area shares, and major deforestation policy programs. We cluster standard errors at the municipality level and our results are robust to Conely spatially-clustered standard errors, as shown in Section 8. The model fit is high overall: the adjusted  $R^2$  of 0.89 reflects the strong predictive role of the lagged outcome and fixed effects, and the within- $R^2$  of 0.29 indicates a meaningful explanatory power for the within-municipality variation.

The results for average effects show that a one-unit increase in the RZSM deficit raises deforestation by 0.00014 (in *asinh* units), significant at the 10 percent level. Although the average annual RZSM deficit across municipalities is about three units, it has a long right tail reaching several hundred units in the most severe grid-years. Hence, the estimated average effect scales up meaningfully during severe drought episodes (e.g., 2015-2016), a point we will



elaborate more in Section 9. This average effect operates mostly at the intensive margin as the extensive-margin effect tends to be much smaller and statistically insignificant at any significance level (Appendix Table E1).

The modest average drought-induced deforestation effect masks substantial heterogeneity in how different types of farms respond to drought stress—the heterogeneity that speaks directly to the economic mechanism of interest. For heterogeneous effects, we separate municipalities by the prevalence of family farming, defined as whether a municipality’s family farm area share lies above or below the 2006 Agricultural Census median. The results are striking. In high-family farm municipalities, a one-unit increase in year-round RZSM deficits raises deforestation by 0.00028, significant at the one percent level. In contrast, the estimated effect turns out to be negative and statistically indistinguishable from zero ( $-0.00005$ ) among low-family-farm municipalities. The difference between the two groups is 0.00032 and precisely estimated ( $p < 0.01$ ). In sum, the deforestation response to agricultural drought is thus as twice as large in high-family-farm municipalities as the average effect, and is entirely absent among low-family-farm municipalities.

While these heterogeneity effects operate primarily at the intensive margin, the extensive-margin heterogeneity effects exhibit a similar pattern. Appendix Table E1 shows that the small and statistically insignificant average effect of agricultural drought on the probability of clearing primary forest is concentrated among high-family-farm municipalities, with the estimated effect being roughly three times as large as that for low-family-farm municipalities. Moreover, only the extensive-margin effect for high-family-farm municipalities is moderately statistically significant.

We examine heterogeneity further by looking at seasonal effects. Table E2 shows that deforestation increases at larger RZSM deficits only during the growing season and among high-family farm municipalities. The deforestation effect is actually negative during the pregrowing season of crop cultivation. All the heterogeneity patterns above point to the center role of family farms in the drought-induced deforestation.

We hypothesize that forest clearing functions as a coping response by family farms to drought-induced agricultural income shocks. Family farms mostly rely on rain-fed production systems and more likely to face binding liquidity constraints, making them more vulnerable to

agricultural drought. On the other hand, larger commercial farms can cope with agricultural drought through irrigated cultivation; they can also draw on wealth or assets to smooth consumption when drought reduces their farm incomes. In Section 7, we examine the impacts of agricultural drought on crop and cattle production, irrigation investments, and rural credit use; the results support our proposed hypothesis.

## 6.2. Agricultural Drought and Environmental Enforcement

A natural concern is that the drought-induced deforestation may reflect a deterioration in monitoring capacity during agricultural drought rather than farmers’ behavioral responses to agricultural drought. If drought disrupts the operations of enforcement agencies—through reduced staffing or reallocation of government resources—observed deforestation could increase as the chance of illegal deforestation being caught becomes lower during agricultural drought. Another concern is that whether the legal enforcement tends to be lower for illegal deforestation that is conducted by family farms, which may contribute to the salient drought-induced deforestation among high-family farm municipalities. We test both hypotheses using data on forest-related environmental fines issued by IBAMA, the Brazilian federal environmental authority.

Table B2 reports the effects of RZSM deficits on fine counts and fine amounts. If drought undermines legal enforcement, we would expect fines to fall during drought years. We find precisely the opposite. Across all Legal Amazon municipalities, a one-unit increase in RZSM deficits raises the count of forest-related fines by 0.0034 (in *asinh* units, significant at the one percent significance level) and their total value by 0.0185 (in *asinh* units, also significant at the one percent significance level). Enforcement, if anything, increases with higher drought intensity, making it unlikely that our deforestation estimates are driven by monitoring failures.

The heterogeneous results further rule out the monitoring-failure story. In high-family-farm municipalities, fine counts rise by 0.0044 ( $p < 0.01$ ) and fine amounts by 0.0222 ( $p < 0.05$ ), mirroring the heterogeneous pattern in deforestation. In low-family-farm municipalities, both estimates are much smaller and statistically insignificant. Fines thus increase where deforestation increases—a pattern that contradicts the monitoring-failure story. The concentration of increased fines among high-family-farm municipalities also rules out weaker legal enforcement

against illegal deforestation by family farms.

As shown in Appendix Table E1, the extensive-margin fine effect mirrors the extensive-margin deforestation effect. On average, a one-unit increase in RZSM deficits raises the probability of any forest-related fine by 0.0013 ( $p < 0.05$ ), and the effect is again larger among high-family-farm municipalities and smaller in low-family-farm municipalities (0.0015 versus 0.0009). Moreover, as with the deforestation response, the enforcement response is concentrated in the growing season and among high-family-farm municipalities.

In sum, both fine counts and incidence rates increase in municipalities where and when forest clearing increases—the opposite of what a monitoring-failure explanation would predict. Putting things together, Tables B1 and B2 establish the core findings of our paper: drought causes deforestation concentrated among high-family-farm municipalities, and this deforestation is both detected by and elicits a response from environmental enforcement agencies. The evidence points to an economic behavioral channel rather than a passive institutional-failure channel. We therefore examine economic mechanisms behind the drought-induced deforestation in the next section.

## 7 — Mechanisms

The results in Section 6 establish that agricultural drought increases deforestation, with this effect concentrated among high-family-farm municipalities, alongside heightened enforcement responses in these same municipalities. This section investigates the agricultural production channel through which drought translates into deforestation pressure. If forest clearing functions as a coping response to agricultural income shocks, drought should reduce agricultural output—particularly in crops that depend on rain-fed production and in municipalities where family farms predominate. Furthermore, high-family-farm municipalities should exhibit limited ability to adapt to agricultural drought, for example, through irrigation investments and rural credit use.

## 7.1. Agricultural Production

We expect that agricultural drought will have adverse impacts on agricultural production, especially for rain-fed crops. We test this prediction by estimating the effect of RZSM deficits on planted areas, harvested areas, output quantities and revenues for each of the four major crops: soybean, sugarcane, cassava, and rice. We focus on these crops, as they dominate agricultural production in the Brazilian Legal Amazon ([Instituto Brasileiro de Geografia e Estatística \(IBGE\) 2023a](#)).<sup>16</sup>

Given the geographic diversity of the Legal Amazon, we define the growing season broadly—from September of the previous year through June of the current year—for summer crops (soybean and rice), encompassing planting, crop development, and harvesting activities across the region, following Rattis et al. ([2021](#)). We also control for pre-growing-season RZSM deficits to account for any early adaptation responses. For multi-year crops (cassava and sugarcane), we use year-round RZSM deficits as the relevant agricultural drought measure.

To facilitate our discussion, we also examine the drought-cattle production effects. In the following, we present results of drought impacts on agricultural production. In particular, we use heterogeneous effects by family-farm prevalence to show that family farms are disproportionately vulnerable to agricultural drought, especially in terms of crop production.

### 7.1.1. Staple Crops: Rice and Cassava

Rice and cassava are the most rain-fed-dependent crops in the Legal Amazon and account for a large share of smallholder farm income ([Instituto Brasileiro de Geografia e Estatística \(IBGE\) 2023a](#)). The adverse impacts of agricultural drought on the production of these two crops provide the clearest evidence of a drought-induced income shock among economically vulnerable producers. We present associated results below.

**Rice.** Table [C1](#) shows the sharpest pattern in the crop production effects. Larger pre-growing-season RZSM deficits significantly decrease planted and harvested areas, output quantities,

16. We exclude first and second corns, another major crops in the Legal Amazon, as they have different growing seasons and we do not have production data by cron types at the municipality-year level. States also dramatically change corn portfolios over time, making it infeasible to conduct meaningful regression analyses.

and revenues in high-family-farm municipalities, with no comparable effect in low-family-farm municipalities. This asymmetry suggests that only family farms significantly reduce crop production in anticipation of early dry soil conditions that would otherwise threaten germination and crop establishment in the later period—likely reflecting their more limited capacity to invest in irrigation or otherwise buffer against anticipated water stress at planting. The significant negative price effect may reflect lower quality of rice produced by family farms.

Furthermore, larger growing-season RZSM deficits significantly decrease planted and harvested areas and output quantities in high-family-farm municipalities (with a similarly negative but statistically insignificant effect on revenues), but significantly increase these same outcomes in low-family-farm municipalities. One potential reason is that rice producers in low-family-farm municipalities can expand irrigated production in response to anticipated higher rice prices—a margin unavailable to predominantly rain-fed family farms. This is consistent with the significant positive effect of growing-season RZSM deficits on rice prices in those municipalities. High-family-farm municipalities experience slightly negative price effects during the growing-season drought, if any.

The negative rice production effects during both seasons for high-family-farm municipalities suggest that rice producers in these municipalities tend to be particularly vulnerable to agricultural drought. This likely reflects family farms’ limited access to irrigation and credit. For instance, smallholder rice producers predominantly rely on rainfall, while large commercial rice producers have access to irrigation. In Section 7.2, we also show that high-family-farm municipalities significantly reduce rural credit use, especially in response to more severe agricultural drought during the growing season. This pronounced concentration of drought-induced rice production losses among high-family-farm municipalities is strongly consistent with our proposed coping-response mechanism: it is precisely the producers most exposed to negative income shocks who have the strongest incentive to supplement lost agricultural revenues through timber extraction and forest clearing.

**Cassava.** Table C2 reports the effects of year-round RZSM deficits on cassava production. Note that cassava is rain-fed in the Legal Amazon, regardless of farm types. Hence, unlike

rice, we do not observe opposite production effects between the two municipality groups for cassava. Higher drought intensity reduces planted and harvested area, output quantities, and revenues in both municipality groups, although there is a positive price effect. Nevertheless, high-family-farm municipalities tend to experience larger negative production effects and smaller positive price effects. Thus, family farms appear more vulnerable to agricultural drought even for cassava production.

### *7.1.2. Cash Crops: Soybean and Sugarcane*

Soybean and sugarcane are the dominant commercial crops in the Brazilian agricultural frontier, typically produced at scale with substantial capital inputs. Like rice, commercial farms use more irrigation for soybean than family farms, although there is no substantial irrigation difference between them for sugarcane. These features help us understand the associated drought-production effects below.

**Soybean.** Table C3 reveals a pattern that is the near-mirror image of the rice effects. Larger RZSM deficits during both the pregrowing and growing seasons significantly decrease planted and harvested areas and output quantities in high-family-farm municipalities, with revenues also falling significantly in response to growing-season drought. In contrast, larger pregrowing-season RZSM deficits significantly increase planted and harvested areas, output quantities, and revenues in low-family-farm municipalities, possibly due to anticipated higher prices, access to irrigation or drought-tolerant soybean seeds. In low-family-farm municipalities, however, larger growing-season RZSM deficits have much smaller and statistically insignificant effects on planted area, harvested area, and output quantities, possibly due to the notably smaller positive price effect. Revenue nonetheless rises significantly, consistent with smaller but consistently positive effects on both quantity and price.

For high-family-farm municipalities, the positive price effect during the pregrowing season largely offsets the negative output quantity effect, making the revenue effect close to null. However, the price effect turns out to be sharply negative during the growing season, leading to an economically sizable and statistically significant negative revenue effect. These asymmetric revenue effects make soybean producers in high-family-farm municipalities more vulnerable

to agricultural drought during the growing season.

**Sugarcane.** Table C4 presents the sugarcane results, which differ from all other crops in an important respect. Larger year-round RZSM deficits have almost no effects on planted and harvested areas as well as output quantities in both high- and low-family-farm municipalities. These null effects may reflect features specific to sugarcane’s perennial, ratoon-based production system, which differs fundamentally from the replanting cycle of the other crops in our sample. Nevertheless, the significant negative price effect, possibly due to lower sugarcane quality, drives a correspondingly significant negative revenue effect in low-family-farm municipalities.

### 7.1.3. *Cattle*

Previous results show that crop production in low-family-farm municipalities is generally less vulnerable to agricultural drought. The only exception is that sugarcane producers in these municipalities are likely to experience significant revenue losses due to lower prices. Here, we show that cattle producers in these municipalities are also likely to suffer from agricultural drought, possibly due to forage deterioration. Table C5 shows that larger year-round RZSM deficits significantly reduce cattle head counts in low-family-farm municipalities. Cattle producers in high-family-farm municipalities experience much smaller and statistically insignificant reductions in head counts. Putting things together, producers in high-family-farm municipalities suffer more from agricultural drought in terms of crop production while producers in low-family-farm municipalities suffer more from agricultural drought in terms of cattle production.

## 7.2. Adaptation

### 7.2.1. *Crop Diversification and Substitution*

Cassava is generally more drought-resilient than rice because it can tolerate relatively long dry spells through deep rooting, reduced leaf area, and flexible growth recovery after rainfall resumes. Rice, in contrast, is more sensitive to water deficits during planting and early growth

stages. For instance, rain-fed upland rice requires adequate soil moisture at planting for reliable germination and seedling establishment. Therefore, there are sizable adverse drought effects for (rain-fed) rice production in high-family-farm municipalities, especially during the pregrowing season, but much smaller negative drought effects for cassava production, as we find in Section 7.1.

For family farms who usually cultivate both crops in the Legal Amazon, we may expect that they either plant more drought-tolerant cassava or harvest more of the existing cassava in response to early-season drought conditions. As shown in Table C6, high-family-farm municipalities producing both rice and cassava significantly harvest more cassava in response to larger pregrowing-season RZSM deficits, although they do not significantly plant more cassava. This finding is consistent with resource-constrained producers adjusting along the lower-cost harvesting margin rather than expanding planted area. Low-family-farm municipalities, on the other hand, significantly plant and harvest more cassava in response to the same drought shock, suggesting that better-resourced producers are able to expand cassava production on both margins simultaneously.

### *7.2.2. Irrigation*

Producers may respond to agricultural drought by expanding irrigation. We obtain irrigation area data for soybean, rice, sugarcane, and other temporary crops from MapBiomass. MapBiomass does not have specific irrigation area data for cassava, consistent with that cassava is mostly rain-fed in the Legal Amazon. For comparison across municipalities, we divide irrigation area by each municipality agricultural establishment area (farm area) to obtain irrigation area share and use it as our outcome variable. Note that soybean and rice are annual crops while sugarcane is a multi-year crop. Considering their difference in the growing cycle, we conduct two sets of regression analyses, one for seasonal drought intensities and one for year-round drought intensities.

For temporary crops like soybean and rice, Table C7 shows that producers may generally decrease irrigation areas in response to more severe drought, specifically during the growing season. This adverse effect tends to be more pronounced in high-family-farm municipalities. Producers in these municipalities also decrease irrigation areas in response to more severe



drought during the pregrowing season, especially for soybean production. In contrast, producers in the low-family-farm municipalities may increase irrigation areas for soybean and rice production to cope with the same seasonal drought shock, although not statistically significant. All these results suggest that family farms may be more vulnerable to agricultural drought as they do not increase but decrease irrigation areas. Table C7 also shows that larger year-round RZSM deficits may decrease irrigation for sugarcane production in both municipality groups, although not statistically significant.

Finally, we also rerun regressions for the drought impacts on crop production by including the lagged crop-specific irrigation area share as an additional control. Results remain almost the same as before, with no changes in the magnitudes and statistical significances of all estimated effects, regardless of municipality groups.<sup>17</sup> Therefore, irrigation plays a limited role in producers' year-over-year responses to agricultural drought.

### 7.2.3. Rural credit

Another commonly-hypothesized risk-coping strategy is rural credit use. In our context, we may expect that producers hit by agricultural drought use more rural credit to smooth consumption or cover production input costs. Theoretically, however, smallholders may not want to take the risk of losing their collateral even when they have access to credit; largeholders may use asset smoothing instead of costly credit. Hence, whether family-farm and commercial-farm producers will use more rural credit or not is an empirical question.

Table C8 shows that on average, producers significantly reduce credit use for crop and cattle production in response to more severe agricultural drought in a year. This negative credit response is concentrated in high-family-farm municipalities, suggesting that family farms generally use less credit in response to agricultural drought. Producers in these municipalities even reduce the use of PRONAF, a family-farm-specific credit line. Producers in low-family-farm municipalities do not experience any significant changes in credit use though. Moreover, producers in both municipality groups do not use more of Proirriga, an irrigation-investment credit line, consistent with our finding that producers do not increase but mostly decrease crop irrigation areas in response to agricultural drought.

17. Results are available upon request.

When we zoom into seasonal patterns, we find that the credit reduction effects mostly apply to the agricultural drought during the growing-season, as shown in Table C9. This seasonal negative credit response is also concentrated in high-family-farm municipalities, with low-family-farm municipalities experiencing almost null effects. Nevertheless, we do find that producers in high-family-farm municipalities significantly use more credit for crop production and use more PRONAF credit in response to more severe drought during the pregrowing season. However, we only see this positive credit effect in terms of credit amounts but not counts, suggesting an intensive-margin effect. Moreover, they significantly reduce credit use for cattle production whereas producers in low-family-farm municipalities increase significantly credit use for cattle production, at least in terms of amounts. Again, producers in both municipality groups do not use more Proirriga credit in response to any seasonal agricultural drought.

In sum, our results suggest that family farms may only have used PRONAF credit to cope with the pregrowing-season agricultural drought and commercial farms may have used rural credit for cattle production in response to the same shock. Both types of farms may not have used Proirriga credit at all to cope with agricultural drought during any seasons. The significant negative impacts of growing-season drought on the credit use for crop and cattle production in high-family-farm municipalities may point to a notable vulnerability of family farms during the growing season.

### 7.3. Synthesis

Our crop production results indicate that more severe agricultural drought tends to hit producers in high-family-farm municipalities more. In particular, they experience significant losses in rice and soybean, while producers in low-family-farm municipalities do not. Although producers in high-family-farm municipalities offset the rice output loss with more cassava harvest during the pregrowing season, they are still exposed to significant crop output losses caused by the growing-season drought. In response to the pregrowing-season drought, they also use more credit for crop production, but they significantly decrease credit use during growing-season drought. In contrast, producers in low-family-farm municipalities generally do not decrease credit use in any season. Taken together, producers in high-family-farm

municipalities may partially cope with agricultural drought during the pregrowing season but not during the growing season. This asymmetric seasonal coping pattern may explain why drought-induced deforestation in high-family-farm municipalities is concentrated in the growing season.

Interestingly, producers in low-family-farm municipalities experience significant drought-induced losses in sugarcane (revenue) and cattle (head count) without responding through deforestation, reflecting their greater capacity to absorb agricultural income shocks through alternative means. For example, they significantly use more credit for cattle in response to more severe drought during the pregrowing season. They also significantly plant and harvest more cassava in response to the same drought shock. When it comes to growing-season drought shock, they significantly expand rice production to capitalize on higher prices. Overall, the asymmetric drought responses between producers from these two municipality groups suggest that family farms are economically more vulnerable to agricultural drought shock, especially during the growing season, which may motivate them to clear forest as a coping strategy.

Nevertheless, producers in both municipality groups do not notably expand irrigation in response to agricultural drought during any season. If anything, producers in high-family-farm municipalities decrease irrigation for all crops considered in this paper, consistent with their limited usage of the irrigation-investment credit line (Proirriga) in response to agricultural drought. Given that, we expect that the drought-induced deforestation will happen repeatedly in high-family-farm municipalities without policy intervention.

## 8 — Robustness Checks

### 8.1. Feedback from deforestation to agricultural drought

Recall that we construct our drought measures using daily root-zone soil moisture (RZSM) data from the Global Land Data Assimilation System (GLDAS). As described in the data section, RZSM is generated by land surface models driven by meteorological forcing data, including precipitation, temperature, and other weather variables. Although the land surface

models rely on static land surface characteristics, the meteorological forcing data vary from year to year. Hence, an important identification threat to our drought-induced deforestation results is the potential feedback from local deforestation to local drought via local rainfall.

Local forest loss can reduce local evapotranspiration, which may in turn reduce local rainfall via moisture recycling (Staal et al. 2020; Smith, Baker, and Spracklen 2023). We use a recent moisture flow dataset ( $1^0$  resolution) from Zhang et al. (2025) to show the contemporary association between local deforestation and local moisture inflow, controlling for grid and time fixed effects as well as prior forest stock. By definition, local moisture inflow refers to moisture that originates from and is absorbed within the same grid. Regression results in Table F1 and Table F2 suggest that larger local deforestation only has a mild and statistically insignificant association with less local moisture inflow, conditional on prior moisture inflow and forest stock. We provide additional consistent evidence using the precipitation data (aggregated at our municipality-RZSM grid level) from TerraClimate. Table F3 shows that larger local deforestation has a limited and statistically insignificant association with less local rainfall, conditional on prior rainfall and forest stock.

We notice that prior forest stock tends to have a much larger association with local moisture inflow and rainfall. This notable association suggests that local forest stock may matter much more than (smaller) local forest loss when it comes to local precipitation. That is why we control for prior forest stock in our main regressions.<sup>18</sup>

## 8.2. Spatial clustering

Our baseline standard errors are clustered by municipality. Because moisture transport and clearing decisions can be spatially correlated across neighboring grids, we also compute Conley standard errors that allow for spatial dependence within a distance cutoff, in addition to serial correlation across years (Appendix Table F4). The drought coefficient and its heterogeneity across family-farm and commercial municipalities remain statistically significant under spatial correction, indicating that the core inference is not an artifact of the clustering choice.

18. Another reason is that more available forest also means larger potential forest loss.

### 8.3. Continuous family-farm area shares

One natural robustness check is to test whether our documented drought-induced deforestation is increasing in the family-farm area share of a municipality. Table F5 shows that drought-induced deforestation is indeed increasing in the municipality-level family-farm area share at a significance level of 10%. As expected, for municipalities with family-farm area shares close to 100%, the drought-induced forestation effect is statistically significant at 5% level and much larger than our main estimate for high-family-farm municipalities reported in Table B1 ( $0.00045 > 0.00028$ ). Accordingly, we also see much larger forest-related environmental fines for these municipalities, although not statistically significant. Therefore, our drought-induced deforestation results largely hold when we directly use municipality-specific family-farm area shares for the heterogeneity analysis.

### 8.4. Rural household incomes

Another natural robustness check is to test whether our documented drought-induced deforestation is increasing in the poor-rural-household share of a municipality. Theoretically, rural poor households tend to clear forest more than non-poor rural households, in response to agricultural drought, as they are more likely to be liquidity- or resource-constrained. We obtain the rural household income distribution data (Tabela 1993) in the 2000 Demographic Census from the Brazilian Institute of Geography and Statistics. This dataset records municipality-level household counts by monthly income bins relative to the minimum monthly wage rate. We classify a rural household with its monthly income less than the minimum wage as a poor household, i.e., for a two-member household, the per-capita monthly income will be less than  $1/2$  of the minimum wage, the income threshold for the federal registration of poor households.

Column (2) of Table F6 shows that the drought-deforestation response is larger and only statistically significant among municipalities with rural poor household shares (number of rural poor households/number of rural households) above the median (40%). We also find that the drought-deforestation response is significantly increasing in the poor-rural-household share of a municipality, although not reported here. These findings are consistent with

our hypothesis that rural poor households tend to clear forest more than non-poor rural households, in response to agricultural drought.

Our data indicate that there is a moderate positive correlation (0.33) between family farm area shares and rural poor household shares at the municipality level. This small correlation allows us to further explore the interactive roles of family farms and rural poor households in drought-induced deforestation. Column (3) shows that low-family-farm municipalities do not deforest more (but less if any) in response to agricultural drought, regardless of rural poor household shares. On the other hand, high-family-farm municipalities respond to drought by significantly deforesting more, regardless of rural poor household shares. As expected, the deforestation response tends to be larger in high-family-farm municipalities with more rural poor households ( $0.00029 > 0.00023$ ), although the difference between more- and less-poor-household municipalities is not statistically significant (not reported here). These findings suggest that both poor and non-poor family-farm households may clear forest in response to agricultural drought. Non-poor family-farm households may be liquidity-constrained as the poor ones, as they may not want to take loans to cope with agricultural drought due to the risk of losing collateral. Therefore, they may choose to clear forest for liquidity as well.

## 8.5. Public forest

As shown in Figure D1, high-family-farm municipalities are mostly located in the northern and eastern parts of the Legal Amazon, whereas low-family-farm municipalities are concentrated in the southern part, the traditional Arc of Deforestation. Although both municipality groups were exposed to severe agricultural drought to some extent, as shown in Figure D2, these two municipality groups may differ in characteristics other than family-farm area shares that could drive drought-induced deforestation. In our regressions, we control for municipality-specific exposure to crop and cattle prices as well as major deforestation regulations. We also examine certain municipality-specific outcomes, such as irrigation and rural credit use, that might capture municipality-specific responses to agricultural drought. However, we may still miss certain factors that could confound our interpretation that family farms drive drought-induced deforestation.

One important factor is forest ownership: relative to private forest, public forest is likely

to be deforested more during drought, unless it is protected, as the opportunity cost of deforesting public forest is close to zero for individual farmers (Pimenta, Reimer, and Novan 2025; Baragwanath and Shinde 2026). We obtain the public land data from the CNFP (Cadastro Nacional de Florestas Públicas, 2024 edition, rectified July 2025). This dataset includes public land (mostly forest) that is either designated (i.e., indigenous territories, conservation units, military land, federal agrarian settlements, etc.) or undesignated (not yet allocated to a specific use). Designated public land is generally protected and held by communities or government agencies, whereas undesignated public land has been claimed by the government but not yet assigned any such status. We therefore focus on undesignated public land given our research purpose.

Column (2) of Table F7 shows that municipalities with public land area shares above the median (0.01) deforest more in response to agricultural drought relative to municipalities with public land area shares below the median ( $0.00014 > 0.00004$ ), although the difference is not statistically significant (p-value of 0.67). Column (1) is the same as that reported in Table B1: municipalities with family farm area shares above the median deforest more in response to agricultural drought while the other municipality group does not ( $0.00028$  versus  $-0.00005$ ), and the difference is statistically significant (p-value  $< 0.01$ ). Thus, family farms tend to play a more important role than public land.

Our data indicate that there is a small positive correlation (0.245) between family farm area shares and public land area shares (public land area/total land area) at the municipality level. This limited correlation allows us to further explore the interactive roles of family farms and public land in drought-induced deforestation. Column (3) shows that low-family-farm municipalities do not deforest more (but less if any) in response to agricultural drought, regardless of public land area shares. On the other hand, high-family-farm municipalities respond to drought by deforesting more, regardless of public land area shares ( $0.00090$  versus  $0.00026$ ), although the estimated effect is only statistically significant for high-family-farm municipalities with public land area shares above the median ( $0.00026$  with p-value  $< 1$ ). These findings suggest that family farms play a central role in drought-induced deforestation while more public land may strengthen that role as it can lower the private opportunity cost of deforestation.

## 8.6. Sensitivity to the 2015–16 megadrought

The El Niño megadrought of 2015–16 is the most extreme event in our sample, and a natural concern is whether the headline result is driven by that single episode. We probe this directly by re-estimating the model excluding 2015–16. The soil-moisture (RZSM) deforestation effect is sensitive to this exclusion: because RZSM deficits load heavily on the 2016 spike, dropping that period sharply attenuates the pooled RZSM coefficient. The meteorological drought measure (SPEI), which draws on broader interannual variation across 2015–17, is more stable and retains a significant positive effect when 2015–16 is excluded. We report both measures transparently rather than selecting the more favorable one (SPEI results reported below): the family-farm concentration—the paper’s central claim—holds under both, but the pooled level effect of soil-moisture drought is disproportionately a megadrought phenomenon, which is itself consistent with the coping interpretation, since the largest income shocks generate the largest clearing responses.

## 8.7. Alternative drought data: SPEI

Our preferred measure is built from daily root-zone soil moisture, which captures the water stress actually experienced by vegetation. As a robustness check we re-estimate the main specifications using the Standardized Precipitation-Evapotranspiration Index (SPEI), a widely used meteorological drought index based on the normalized difference between precipitation and potential evapotranspiration (Vicente-Serrano, Beguería, and López-Moreno 2010; Reig-Gracia and Latorre Garcés 2023). To align with the resolution of our RZSM-based drought measure, we obtain the 0.25-degree SPEI data from Liu et al. (2024). Because SPEI is conventionally computed at a monthly resolution, we use monthly SPEI to calculate year-round drought intensity, following the same procedure as our daily-RZSM-based year-round drought intensity: we calculate monthly SPEI deficits relative to the bottom 10th percentile threshold of -1.28, namely  $(-1.28 - SPEI) \times I_{\{SPEI < -1.28\}}$ , and then sum across the 12 months of a year.

As shown in Table F8, the two drought measures agree on the central result—more intense drought raises deforestation—but differ in other aspects. Under SPEI the pooled



deforestation response is highly statistically significant, but the difference between high- and low-family-farm municipalities is not. Moreover, SPEI yields no statistically significant effects on forest-related environmental fines in either the pooled or heterogeneous specifications. The fine amount and count response patterns are not consistent as well: more fine counts do not lead to more fine amounts. This stands in contrast to our preferred RZSM-based measure, which detects strong and statistically significant enforcement effects with consistent heterogeneous patterns.

The divergence described above is consistent with SPEI’s coarser, atmosphere-based signal being less able to capture the localized soil-water stress that plausibly drives both deforestation and the enforcement response to it, whereas RZSM, an agriculturally grounded measure of root-zone water availability, appears better suited to detecting this link. We acknowledge that the RZSM- and SPEI-based drought measures may capture different types of drought—agricultural and meteorological drought, respectively. Nevertheless, that the main deforestation result survives both strengthens our confidence that it reflects genuine water stress rather than the idiosyncrasies of a single index.

## 9 — Policy Implications

Our estimates carry a direct policy reading: because drought-induced clearing is likely to be a coping response to a negative liquidity shock, the associated carbon externality is *avoidable* through instruments that relax the constraint. We make this concrete in three steps, deferring the full accounting to Appendix G.

**Drought-attributable forest loss.** To quantify how much forest is lost to drought, we back out the drought component of clearing from the estimated dynamic model, as reported in Table B1. We simulate each municipality-RZSM grid’s asinh clearing path twice—once with its realized RZSM deficits and once setting all deficits to zero—and sum the back-transformed difference over 2006–2019.<sup>19</sup> Based on the average-effect model, drought accounts for roughly 257 square kilometers of forest loss relative to the year-2000 forest baseline, or about 1,836

19. The difference in forest stock for the initial year 2005 is mechanically zero in our dynamic model.

hectares per year. The estimate is robust to alternative forest base years (1985, 1995, and 2000 all deliver similar totals), reflecting that drought primarily accelerates clearing of older, higher-value forest rather than young regrown forest. Although this is a small share of total Legal-Amazon clearing during 2006-2019 ( $\approx 0.15\%$ ), it is highly concentrated: in the 2015–16 megadrought, drought-attributable loss reached over 4.4 times the annual average.

**Committed carbon and its social cost.** Converting hectares to committed emissions using an aboveground-biomass emission factor of approximately 300 tons of CO<sub>2</sub> per hectare (Ometto et al. 2023; IPCC 2019), drought-induced clearing commits on the order of 0.55 million tons of CO<sub>2</sub> per year. Valued at a central social cost of carbon of \$190 per ton (U.S. Environmental Protection Agency 2023), this implies a committed carbon damage of roughly \$105 million per year on average, rising to about \$463 million in the 2015–16 megadrought years; the sensitivity of this figure to the biomass and social-cost-of-carbon assumptions is reported in Appendix G (Interagency Working Group on Social Cost of Greenhouse Gases 2021; Nordhaus 2017; Barrage and Nordhaus 2024). Because our emission factor counts aboveground live biomass only, and because our counterfactual holds constant the many other drivers of clearing, these figures should be read as a conservative, indicative lower bound on the climate externality of Amazon drought operating through the agricultural channel.

**Policy leverage: rural finance and public-land protection.** The central policy implication follows directly from the mechanism we identify rather than the magnitude of the estimated effects. Enforcement and market-based conservation policies—the workhorses of Amazon forest protection—are designed to discourage profitable forest conversion. They are less effective when deforestation is driven by temporary liquidity needs following adverse climate shocks. Our falsification exercise reinforces this interpretation: environmental fines increase during drought years, implying that additional clearing occurs despite, rather than because of, weaker monitoring. Simply intensifying enforcement is therefore unlikely to eliminate the coping motive underlying drought-induced deforestation.

Policies that relax short-run financial constraints may instead target the mechanism directly. Expanding access to rural credit, crop insurance, and drought-indexed transfers

can reduce the need for liquidity-constrained households to finance temporary income losses by liquidating forest assets. In Brazil, however, access to these instruments remains highly unequal. Despite PRONAF being the country’s principal rural credit program, only a small minority of family farmers receive formal credit, with access particularly limited among smaller producers and in the North, where climate vulnerability is greatest (Souza and Albuquerque 2023). Recent evidence further shows that expanding technical assistance (ATER) and strengthening producer cooperatives substantially increase access to PRONAF, suggesting that financial inclusion depends not only on the availability of subsidized credit but also on complementary rural institutions that lower informational and administrative barriers to borrowing (Leal, Assunção, and Hoover 2026). Rather than viewing rural finance solely as a development or social protection policy, our results suggest that improving access to financial services for liquidity-constrained households can also complement climate mitigation by reducing the incentive to clear forest following temporary income shocks.

Our results further highlight the importance of public-land governance. The public-forest results in Section 8 show that drought-induced deforestation is concentrated in municipalities with both a high prevalence of family farming and above-median shares of undesignated public land. This pattern suggests that weakly protected public forests provide the primary margin of adjustment when financially constrained households respond to climate shocks. Strengthening the protection and formal designation of undesignated public lands can therefore complement rural finance policies by limiting opportunities for coping-driven encroachment onto low-opportunity-cost forest.

More broadly, our findings suggest that financial resilience and forest protection should be viewed as complementary components of climate policy rather than independent objectives. Expanding access to rural finance, insurance, and extension services can reduce the incentive to liquidate forest following adverse weather shocks, while stronger protection of undesignated public lands limits the availability of forest as a coping asset. Together, these policies address both the demand for forest clearing created by household liquidity constraints and the supply of vulnerable public forest available for conversion.

## 10 — Conclusion

This paper studies whether and how agricultural drought leads to deforestation in the Brazilian Legal Amazon where family and commercial farms coexist. Using high-resolution root-zone soil moisture and land cover data, we find that increases in soil moisture deficits lead to higher deforestation, with no statistically detectable within-year feedback from local deforestation to local moisture inflow or rainfall once we condition on prior forest stock. Importantly, the deforestation response is overwhelmingly concentrated where family farming predominates, although both family-farm-predominant and commercial-farm-predominant municipalities experience significant crop or livestock losses. Furthermore, producers in high-family-farm municipalities respond to more severe agricultural drought by reducing, rather than increasing, crop production, suggesting that their deforestation response is likely a signature of binding liquidity constraints rather than of profitability. Finally, these producers are also severely constrained in terms of adaptation: they neither invest more in irrigation nor use more rural credit, suggesting an urgent need for policy intervention.

These findings carry a distinct policy message. The dominant instruments of Amazon conservation—command-and-control enforcement and market-based incentives—are built for stable economic conditions and do not reach the coping motive that drives clearing when smallholder agricultural income collapses. Because drought-induced clearing is likely a response to a liquidity shock, it is avoidable through instruments that relax the constraint: affordable or accessible rural credit, crop insurance, or drought-indexed transfers. Our carbon accounting indicates that the stakes are material—drought-attributable clearing commits on more than half a million tons of CO<sub>2</sub> per year, worth about \$105 million per year at a central social cost of carbon and far more in megadrought years, suggesting substantial room for such policy interventions. Overall, our results suggest that climate adaptation for smallholders and climate mitigation for the planet, in the Brazilian Amazon, are complements rather than competing claims on the policy budget.

## References

- Abatzoglou, John T., Solomon Z. Dobrowski, Sean A. Parks, and Katherine C. Hegewisch. 2018. “TerraClimate, a High-Resolution Global Dataset of Monthly Climate and Climatic Water Balance from 1958–2015.” *Scientific Data* 5:170191. <https://doi.org/10.1038/sdata.2017.191>. (Cited on page 18).
- Abman, Ryan, and Conor Carney. 2020. “Agricultural productivity and deforestation: Evidence from input subsidies and ethnic favoritism in Malawi.” *Journal of Environmental Economics and Management* 103:102342. (Cited on page 3).
- Amacher, Gregory S., Erkki Koskela, and Markku Ollikainen. 2009. “Deforestation and land use under insecure property rights.” *Environment and Development Economics* 14 (3): 281–303. (Cited on page 3).
- Amacher, Gregory S., Markku Ollikainen, and Erkki Koskela. 2009. *Economics of Forest Resources*. Cambridge, MA: MIT Press. (Cited on pages 2, 11).
- Anderson, Liana O., Germano Ribeiro Neto, Ana Paula Cunha, Marisa Gesteira Fonseca, Yhasmin Mendes de Moura, Ricardo Dalagnol, Fabien Hubert Wagner, and Luiz Eduardo Oliveira e Cruz de Aragão. 2018. “Vulnerability of Amazonian forests to repeated droughts.” *Philosophical Transactions of the Royal Society B* 373 (1760): 20170411. (Cited on pages 2, 10).
- Aragão, Luiz E. O. C., Liana O. Anderson, Marisa G. Fonseca, Thais M. Rosan, Laura B. Vedovato, Fabien H. Wagner, Camila V. J. Silva, et al. 2018. “21st Century drought-related fires counteract the decline of Amazon deforestation.” *Nature Communications* 9:536. (Cited on page 10).
- Araujo, Rafael, Juliano Assunção, Marina Hirota, and José A. Scheinkman. 2023. “Estimating the spatial amplification of damage caused by degradation in the Amazon.” *Proceedings of the National Academy of Sciences* 120 (46): e2312451120. (Cited on page 2).
- Araujo, Rafael, Francisco Costa, and Marcelo Sant’Anna. 2025. “Efficient Conservation of the Brazilian Amazon: Estimates from a Dynamic Model.” Forthcoming. doi:10.1093/restud/rdaf031, *The Review of Economic Studies*, (cited on pages 2, 7).
- Assunção, Juliano, Clarissa Gandour, and Romero Rocha. 2023. “DETER-ing Deforestation in the Amazon: Environmental Monitoring and Law Enforcement.” *American Economic Journal: Applied Economics* 15 (2): 125–156. (Cited on pages 2, 4).

- Assunção, Juliano, Clarissa Gandour, Romero Rocha, and Rudi Rocha.** 2020. “The Effect of Rural Credit on Deforestation: Evidence from the Brazilian Amazon.” *The Economic Journal* 130 (626): 290–330. (Cited on pages 2, 4, 7, 8, 9, 21, 22, 67).
- Assunção, Juliano, Clarissa Gandour, and Rudi Rocha.** 2015. “Deforestation slowdown in the Brazilian Amazon: prices or policies?” *Environment and Development Economics* 20 (6): 697–722. (Cited on pages 8, 9, 17, 22).
- Assunção, Juliano, Robert McMillan, Joshua Murphy, and Eduardo Souza-Rodrigues.** 2023. “Optimal Environmental Targeting in the Amazon Rainforest.” *The Review of Economic Studies* 90 (4): 1608–1641. (Cited on page 7).
- Assunção, Juliano, and Romero Rocha.** 2019. “Getting greener by going black: the effect of blacklisting municipalities on Amazon deforestation.” *Environment and Development Economics* 24 (2): 115–137. (Cited on pages 4, 9).
- Baragwanath, Kathryn, and Ella Bayi.** 2020. “Collective property rights reduce deforestation in the Brazilian Amazon.” *Proceedings of the National Academy of Sciences* 117 (34): 20495–20502. (Cited on pages 2, 4, 8).
- Baragwanath, Kathryn, and Nilesh Shinde.** 2025. “Beyond the canopy: How satellite data detection thresholds influence policy evaluation and deforestation behavior.” *Journal of Environmental Economics and Management*, 103219. (Cited on pages 2, 8, 9).
- . 2026. *Clearing to Claim: The Perverse Incentives of Land Regularization in the Brazilian Amazon*. Working Paper. Ronald Coase Institute. (Cited on pages 4, 39).
- Barrage, Lint, and William D. Nordhaus.** 2024. “Policies, Projections, and the Social Cost of Carbon: Results from the DICE-2023 Model.” *Proceedings of the National Academy of Sciences* 121 (13): e2312030121. <https://doi.org/10.1073/pnas.2312030121>. (Cited on pages 42, 81).
- Carreira, Igor, Francisco Costa, and João Paulo Pessoa.** 2024. “The deforestation effects of trade and agricultural productivity in Brazil.” *Journal of Development Economics* 167:103217. (Cited on page 3).
- Central Bank of Brazil (BCB).** 2023. *Microdata from the Rural Credit and Proagro Operations System (SICOR)*. <https://www.bcb.gov.br/en/statistics/ruralcredit>. Accessed 2025-04-12. Temporal coverage 2013–2023-09. (Cited on page 21).
- Chen, Yang, Joanne Hall, Dave van Wees, Niels Andela, Stijn Hantson, Louis Giglio, Guido R. van der Werf, Douglas C. Morton, and James T. Randerson.** 2023. “Multi-decadal trends and variability in burned area from the fifth version of the Global Fire Emissions Database (GFED5).” *Earth System Science Data* 15 (11): 5227–5259. (Cited on page 2).

- Chomitz, Kenneth M., and David A. Gray.** 1996. “Roads, Land Use, and Deforestation: A Spatial Model Applied to Belize.” *The World Bank Economic Review* 10 (3): 487–512. (Cited on pages 2, 11).
- Cisneros, Elías, Sophie Lian Zhou, and Jan Börner.** 2015. “Naming and shaming for conservation: evidence from the Brazilian Amazon.” *PloS one* 10 (9): e0136402. (Cited on pages 4, 9).
- De Janvry, Alain, Marcel Fafchamps, and Elisabeth Sadoulet.** 1991. “Peasant household behaviour with missing markets: some paradoxes explained.” *The economic journal* 101 (409): 1400–1417. (Cited on pages 2, 11).
- Desbureaux, Sébastien, and Richard Damania.** 2018. “Rain, forests and farmers: Evidence of drought induced deforestation in Madagascar and its consequences for biodiversity conservation.” *Biological Conservation* 221:357–364. (Cited on page 3).
- Fafchamps, Marcel, Christopher Udry, and Katherine Czukas.** 1998. “Drought and Saving in West Africa: Are Livestock a Buffer Stock?” *Journal of Development Economics* 55 (2): 273–305. (Cited on page 3).
- Ferraro, Paul J., and Rhita Simorangkir.** 2020. “Conditional cash transfers to alleviate poverty also reduced deforestation in Indonesia.” *Science Advances* 6 (24): eaaz1298. (Cited on pages 3, 7).
- Gibbs, Holly K, Jacob Munger, Jessica L’Roe, Paulo Barreto, Ritaumaria Pereira, Matthew Christie, Ticiana Amaral, and Nathalie F Walker.** 2016. “Did ranchers and slaughterhouses respond to zero-deforestation agreements in the Brazilian Amazon?” *Conservation letters* 9 (1): 32–42. (Cited on pages 2, 4, 9).
- Hanusch, Marek.** 2023. *A balancing act for Brazil’s Amazonian states: An economic memorandum*. World Bank Publications. (Cited on page 2).
- He, Xi, and Zhenshan Chen.** 2022. “Weather, Cropland Expansion, and Deforestation in Ethiopia.” *Journal of Environmental Economics and Management* 111:102586. ISSN: 0095-0696. <https://doi.org/10.1016/j.jeem.2021.102586>. (Cited on page 3).
- Heilmayr, Robert, Lisa L Rausch, Jacob Munger, and Holly K Gibbs.** 2020. “Brazil’s Amazon soy moratorium reduced deforestation.” *Nature Food* 1 (12): 801–810. (Cited on pages 2, 4, 9).
- Hsiao, Allan, Jacob Moscona, Benjamin A. Olken, and Karthik A. Sastry.** 2026. *Climate Change, Deforestation, and the Expansion of the Global Agricultural Frontier*. NBER Working Paper 35029. National Bureau of Economic Research. (Cited on pages 2, 3, 7, 12).



- Hultgren, Andrew, Tamma Carleton, Michael Delgado, Diana R. Gergel, Michael Greenstone, Trevor Houser, Solomon Hsiang, Amir Jina, Robert E. Kopp, Steven B. Malevich, et al. 2025. “Impacts of climate change on global agriculture accounting for adaptation.” *Nature* 642 (8068): 644–652. (Cited on page 7).
- Instituto Brasileiro de Geografia e Estatística (IBGE). 2006. *Censo Agropecuário 2006*. IBGE, Rio de Janeiro. Accessed 2025-04-12. (Cited on pages 9, 10, 20).
- . 2023a. *Pesquisa Agrícola Municipal (PAM)*. <https://www.ibge.gov.br/estatisticas/economicas/agricultura-e-pecuaria/9109-producao-agricola-municipal-culturas-temporarias-e-permanentes.html>. Accessed 2025-04-12. (Cited on pages 20, 28).
- . 2023b. *Pesquisa Pecuária Municipal (PPM)*. <https://www.ibge.gov.br/estatisticas/economicas/agricultura-e-pecuaria/9107-producao-da-pecuaria-municipal.html>. Accessed 2025-04-12. (Cited on page 21).
- Interagency Working Group on Social Cost of Greenhouse Gases. 2021. *Technical Support Document: Social Cost of Carbon, Methane, and Nitrous Oxide—Interim Estimates under Executive Order 13990*. Technical report. Washington, DC: U.S. Government. (Cited on pages 42, 81).
- IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4: Agriculture, Forestry and Other Land Use*. Technical report. Geneva: Intergovernmental Panel on Climate Change. (Cited on pages 42, 80).
- Jayachandran, Seema. 2013. “Liquidity Constraints and Deforestation: The Limitations of Payments for Ecosystem Services.” *American Economic Review* 103 (3): 309–313. (Cited on pages 3, 7, 21).
- Khan, Bhoktear, Piyush Mehta, Dongyang Wei, Hanan Abou Ali, Oluseun Adeluyi, Tunrayo Alabi, Olawale Olayide, John Uponi, and Kyle Frankel Davis. 2025. “Cropland Expansion Links Climate Extremes and Diets in Nigeria.” *Science Advances* 11, no. 2 (January): eado5541. ISSN: 2375-2548. <https://doi.org/10.1126/sciadv.ado5541>. (Cited on pages 3, 7).
- Klosin, Sylvia. 2024. “Dynamic biases of static panel data estimators.” *arXiv preprint arXiv:2410.16112*, (cited on page 14).
- Leal, Alan, Juliano Assunção, and Natalie Hoover. 2026. *Closing the Rural Credit Divide: Pathways to Increase PRONAF Access for Smallholders*. Technical report. Policy report. Rio de Janeiro, Brazil: Climate Policy Initiative, April. <https://www.climatepolicyinitiative.org/publication/closing-the-rural-credit-divide-pathways-to-increase-pronaf-access-for-smallholders/>. (Cited on pages 9, 43).
- Leblois, Antoine. 2021. “Mitigating the impact of bad rainy seasons in poor agricultural regions to tackle deforestation.” *Environmental Research Letters* 16 (3): 034015. (Cited on page 3).



- Liu, Xuebang, Shuying Yu, Zhiwei Yang, Jianquan Dong, and Jian Peng.** 2024. “The first global multi-timescale daily SPEI dataset from 1982 to 2021.” *Scientific Data* 11 (1): 223. (Cited on pages 40, 78).
- MapBiomass Project.** 2024. *MapBiomass Project — Collection 9 of the Annual Land Use Land Cover Maps of Brazil*. <https://mapbiomas.org/en>. Accessed 2025-04-12. (Cited on pages 8, 18).
- Nordhaus, William D.** 2017. “Revisiting the Social Cost of Carbon.” *Proceedings of the National Academy of Sciences* 114 (7): 1518–1523. <https://doi.org/10.1073/pnas.1609244114>. (Cited on pages 42, 81).
- Ometto, Jean P., et al.** 2023. “L-band Synthetic Aperture Radar and LiDAR Estimates of Aboveground Biomass in the Brazilian Amazon.” VOLUME/ARTICLE NUMBER AND FULL AUTHOR LIST NEEDED, *Scientific Data* 10. (Cited on pages 8, 42, 80).
- Ordóñez, Pablo J., Protensia Hadunka, Gemma Del Rossi, and Kathy Baylis.** 2025. “When crops fail, forests follow: Agricultural shocks and deforestation in Zambia.” *Proceedings of the National Academy of Sciences* 122 (40): e2427156122. (Cited on page 3).
- Osman, Mahmoud, Benjamin Zaitchik, Jason Otkin, and Martha Anderson.** 2024. “A global flash drought inventory based on soil moisture volatility.” *Scientific Data* 11 (1): 965. <https://doi.org/10.1038/s41597-024-03809-9>. (Cited on pages 4, 5, 10, 16).
- Otkin, Jason A., Mark Svoboda, Eric D. Hunt, Trent W. Ford, Martha C. Anderson, Christopher Hain, and Jeffrey B. Basara.** 2018. “Flash Droughts: A Review and Assessment of the Challenges Imposed by Rapid-Onset Droughts in the United States.” *Bulletin of the American Meteorological Society* 99 (5): 911–919. <https://doi.org/10.1175/BAMS-D-17-0149.1>. (Cited on page 10).
- Papastefanou, Phillip, Thomas AM Pugh, Allan Buras, Katrin Fleischer, Thorsten EE Grams, Thomas Hickler, David Lapola, Daijun Liu, Christian S Zang, and Anja Rammig.** 2024. “Simulated sensitivity of the Amazon rainforest to extreme drought.” *Environmental Research Letters* 19 (12): 124072. (Cited on page 2).
- Pattanayak, Subhrendu K., and Erin O. Sills.** 2001. “Do Tropical Forests Provide Natural Insurance? The Microeconomics of Non-Timber Forest Product Collection in the Brazilian Amazon.” *Land Economics* 77 (4): 595–612. (Cited on page 3).
- Pfaff, Alexander S. P.** 1999. “What Drives Deforestation in the Brazilian Amazon? Evidence from Satellite and Socioeconomic Data.” *Journal of Environmental Economics and Management* 37 (1): 26–43. (Cited on pages 2, 3, 11).
- Pimenta, Bruno, Matthew Reimer, and Kevin Novan.** 2025. “The Differential Responses of Farmers on Private and Public Lands to Droughts in the Brazilian Amazon.” *Working Paper*, (cited on pages 7, 39).

- Pindyck, Robert S.** 2019. “The Social Cost of Carbon Revisited.” *Journal of Environmental Economics and Management* 94:140–160. <https://doi.org/10.1016/j.jeem.2019.02.003>. (Cited on page 81).
- Rattis, Ludmila, Paulo M. Brando, Marcia N. Macedo, Stephanie A. Spera, Andrea D. A. Castanho, Eduardo Q. Marques, Nathane Q. Costa, Divino V. Silverio, and Michael T. Coe.** 2021. “Climatic limit for agriculture in Brazil.” *Nature Climate Change* 11 (12): 1098–1104. (Cited on pages 17, 28).
- Reig-Gracia, Fergus, and Borja Latorre Garcés.** 2023. *SPEIbase v.2.9 [Dataset]*. <https://digital.csic.es/handle/10261/308758>. V. 2.9. <https://doi.org/10.20350/digitalCSIC/15470>. (Cited on pages 17, 40).
- Rosenzweig, Mark R., and Kenneth I. Wolpin.** 1993. “Credit Market Constraints, Consumption Smoothing, and the Accumulation of Durable Production Assets in Low-Income Countries: Investments in Bullocks in India.” *Journal of Political Economy* 101 (2): 223–244. (Cited on page 3).
- Rudel, Thomas K., Oliver T. Coomes, Emilio Moran, Frédéric Achard, Arild Angelsen, Jianchu Xu, and Eric F. Lambin.** 2005. “Forest Transitions: Towards a Global Understanding of Land Use Change.” *Global Environmental Change* 15 (1): 23–31. ISSN: 0959-3780. <https://doi.org/10.1016/j.gloenvcha.2004.11.001>. (Cited on pages 2, 11).
- Smith, Callum, Jessica C. A. Baker, and Dominick V. Spracklen.** 2023. “Tropical deforestation causes large reductions in observed precipitation.” *Nature* 615 (7951): 270–275. (Cited on pages 2, 14, 18, 36).
- Souza, Priscila, and Amanda de Albuquerque.** 2023. *Family Farming in Brazil: Inequalities in Credit Access*. Technical report. Policy report. Rio de Janeiro, Brazil: Climate Policy Initiative, November. <https://www.climatepolicyinitiative.org/publication/family-farming-in-brazil-inequalities-in-credit-access/>. (Cited on pages 9, 43).
- Souza, Priscila, Stela Herschmann, and Juliano Assunção.** 2020. *Rural Credit Policy in Brazil: Agriculture, Environmental Protection, and Economic Development*. Technical report. Rio de Janeiro: Climate Policy Initiative, December. <https://climatepolicyinitiative.org>. (Cited on page 9).
- Souza Jr., Carlos M., Julia Z. Shimbo, Marcos R. Rosa, Leandro L. Parente, Ane A. Alencar, Bernardo F. T. Rudorff, Heinrich Hasenack, Marcelo Matsumoto, Laerte G. Ferreira, Pedro W. M. Souza-Filho, et al.** 2020. “Reconstructing Three Decades of Land Use and Land Cover Changes in Brazilian Biomes with Landsat Archive and Earth Engine.” *Remote Sensing* 12 (17): 2735. <https://doi.org/10.3390/rs12172735>. (Cited on pages 8, 18).

- Souza-Rodrigues, Eduardo.** 2019. “Deforestation in the Amazon: A Unified Framework for Estimation and Policy Analysis.” *The Review of Economic Studies* 86 (6): 2713–2744. (Cited on pages 2, 4, 8).
- Staal, Arie, Bernardo M. Flores, Ana Paula D. Aguiar, Joyce H. C. Bosmans, Ingo Fetzer, and Obbe A. Tuinenburg.** 2020. “Feedback between drought and deforestation in the Amazon.” *Environmental Research Letters* 15 (4): 044024. (Cited on pages 2, 5, 7, 14, 18, 36).
- Szerman, Dimitri, Juliano Assunção, Molly Lipscomb, and Ahmed Mushfiq Mo-barak.** 2024. *Electricity, Agricultural Productivity, and Deforestation: Evidence from Brazil*. Technical report. Working paper. (Cited on page 3).
- Tabe-Ojong, Martin Paul Jr., Emmanuel Tolani, and Ernest L. Molua.** 2026. “Rolling Back the Tides: Impact of Droughts on Crop Diversification and Cropland Expansion.” *Journal of Environmental Economics and Management* 136 (February): 103260. ISSN: 0095-0696. <https://doi.org/10.1016/j.jeem.2025.103260>. (Cited on pages 3, 7).
- U.S. Environmental Protection Agency.** 2023. *Report on the Social Cost of Greenhouse Gases: Estimates Incorporating Recent Scientific Advances*. Technical report. Docket ID No. EPA-HQ-OAR-2021-0317. Washington, DC: U.S. Environmental Protection Agency, National Center for Environmental Economics. (Cited on pages 42, 80, 81).
- Vicente-Serrano, Sergio M., Santiago Beguería, and Juan I. López-Moreno.** 2010. “A Multiscalar Drought Index Sensitive to Global Warming: The Standardized Precipitation Evapotranspiration Index.” *Journal of Climate* 23 (7): 1696–1718. <https://doi.org/10.1175/2009JCLI2909.1>. (Cited on pages 17, 40).
- West, Thales AP, and Philip M Fearnside.** 2021. “Brazil’s conservation reform and the reduction of deforestation in Amazonia.” *Land use policy* 100:105072. (Cited on pages 8, 15).
- Willhite, Donald A., and Michael H. Glantz.** 1985. “Understanding the Drought Phenomenon: The Role of Definitions.” *Water International* 10 (3): 111–120. (Cited on pages 5, 16).
- Wunder, Sven, Jan Börner, Gerald Shively, and Miriam Wyman.** 2014. “Safety Nets, Gap Filling and Forests: A Global-Comparative Perspective.” *World Development* 64, no. Supplement 1 (December): S29–S42. ISSN: 0305-750X. <https://doi.org/10.1016/j.worlddev.2014.03.005>. (Cited on page 3).
- You, Nanshan, Jessica Till, David B. Lobell, Peng Zhu, Paul C. West, Hui Kong, Wei Li, et al.** 2025. “Climate-Driven Global Cropland Changes and Consequent Feedbacks.” *Nature Geoscience* 18, no. 7 (July): 639–645. ISSN: 1752-0894. <https://doi.org/10.1038/s41561-025-01724-1>. (Cited on page 3).

- Yuan, Xing, Yumiao Wang, Peng Ji, Peili Wu, Justin Sheffield, and Jason A. Otkin.** 2023. “A global transition to flash droughts under climate change.” *Science* 380 (6641): 187–191. <https://doi.org/10.1126/science.abn6301>. (Cited on page 10).
- Zaveri, Esha, Jason Russ, and Richard Damania.** 2020. “Rainfall anomalies are a significant driver of cropland expansion.” *Proceedings of the National Academy of Sciences* 117 (19): 10225–10233. (Cited on page 3).
- Zhang, Yu, Deyu Zhong, Rongrong Cai, Mingxi Zhang, Yinglin Tian, Di Xie, Yuan Cao, and Yuantao Mei.** 2025. “A multi-model based dataset of global atmospheric moisture source-sink relationships and atmospheric basins.” *Scientific Data* 12 (1): 1842. (Cited on pages 15, 18, 36, 73).

# A — Summary Statistics Tables

**Table A1 — Summary Stats of Main Variables**

|                                                                 | N      | Mean         | SD            | Min    | Max         |
|-----------------------------------------------------------------|--------|--------------|---------------|--------|-------------|
| <i>Panel A. Municipality-RZSM grid-by-year data (2005-2019)</i> |        |              |               |        |             |
| loss2000 (km2)                                                  | 197400 | 0.9766       | 3.0578        | 0.0000 | 84.9930     |
| loss2000_dummy                                                  | 197400 | 0.6402       | 0.4799        | 0.0000 | 1.0000      |
| loss2000_asinh                                                  | 197400 | 0.4313       | 0.8028        | 0.0000 | 5.1358      |
| RZSM_dintensity                                                 | 197400 | 2.9662       | 17.5641       | 0.0000 | 548.1848    |
| RZSM_ddays                                                      | 197400 | 2.0411       | 6.9667        | 0.0000 | 86.0000     |
| RZSM_drought_dummy                                              | 197400 | 0.1530       | 0.3599        | 0.0000 | 1.0000      |
| stock (km2)                                                     | 197400 | 237.2575     | 260.8738      | 0.0000 | 769.3430    |
| stock_dummy                                                     | 197400 | 0.9810       | 0.1366        | 0.0000 | 1.0000      |
| stock_asinh                                                     | 197400 | 4.9013       | 2.1419        | 0.0000 | 7.3387      |
| <i>Panel B. Municipality-by-year data (2005-2019)</i>           |        |              |               |        |             |
| fine_dummy                                                      | 11565  | 0.4952       | 0.5000        | 0.0000 | 1.0000      |
| fines_count                                                     | 11565  | 7.2472       | 25.0234       | 0.0000 | 841.0000    |
| lfines_count_asinh                                              | 11565  | 1.1922       | 1.5057        | 0.0000 | 7.4277      |
| fines_amount (R\$)                                              | 11565  | 2868637.6677 | 21684406.0964 | 0.0000 | 1.4455e+09  |
| lfines_amount_asinh                                             | 11565  | 6.5638       | 6.9223        | 0.0000 | 21.7849     |
| muni_RZSM_dintensity                                            | 11565  | 2.3849       | 10.6657       | 0.0000 | 204.2956    |
| muni_RZSM_ddays                                                 | 11565  | 1.6634       | 5.3052        | 0.0000 | 47.4442     |
| muni_drought_dummy                                              | 11565  | 0.2610       | 0.4392        | 0.0000 | 1.0000      |
| forest_stock (km2)                                              | 11565  | 4054.3146    | 11433.9964    | 2.5911 | 144497.1406 |
| forest_stock_log                                                | 11565  | 7.2298       | 1.8440        | 1.6805 | 12.5742     |
| <i>Panel C. Family farm data (2006)</i>                         |        |              |               |        |             |
| familyfarm_area (hectares)                                      | 770    | 33051.31     | 36112.35      | 110    | 247487      |
| farmarea_tot (hectares)                                         | 770    | 150213.2     | 190151.6      | 756    | 1581759     |
| familyfarm_areashare                                            | 770    | 33.2707      | 21.3532       | 0.0785 | 97.0252     |

**Table A2 — Summary Stats of Major Crop Production**

|                                                                                  | N     | Mean        | SD          | Min      | Max          |
|----------------------------------------------------------------------------------|-------|-------------|-------------|----------|--------------|
| <i>Panel A. Municipality-by-year crop production data: Soybean (2005-2019)</i>   |       |             |             |          |              |
| area_planted (hectares)                                                          | 2505  | 51824.4056  | 86732.2096  | 60.0000  | 635000.0000  |
| area_planted_log                                                                 | 2505  | 9.6412      | 1.7521      | 4.0943   | 13.3614      |
| area_harvested (hectares)                                                        | 2505  | 51753.7242  | 86581.2852  | 60.0000  | 630000.0000  |
| area_harvested_log                                                               | 2505  | 9.6396      | 1.7526      | 4.0943   | 13.3535      |
| quantity (metric tons)                                                           | 2505  | 158126.6930 | 272845.8923 | 162.0000 | 2232000.0000 |
| quantity_log                                                                     | 2505  | 10.7048     | 1.7937      | 5.0876   | 14.6184      |
| revenue (1000 R\$)                                                               | 2505  | 120385.4371 | 220626.0448 | 103.0000 | 2165794.0000 |
| revenue_log                                                                      | 2505  | 10.3229     | 1.8918      | 4.6347   | 14.5883      |
| <i>Panel B. Municipality-by-year crop production data: Sugarcane (2005-2019)</i> |       |             |             |          |              |
| area_planted (hectares)                                                          | 2670  | 1516.6813   | 5863.1935   | 1.0000   | 53861.0000   |
| area_planted_log                                                                 | 2670  | 3.9095      | 2.1728      | 0.0000   | 10.8942      |
| area_harvested (hectares)                                                        | 2670  | 1505.9101   | 5820.4349   | 1.0000   | 53861.0000   |
| area_harvested_log                                                               | 2670  | 3.8898      | 2.1741      | 0.0000   | 10.8942      |
| quantity (metric tons)                                                           | 2670  | 104659.9273 | 425359.1117 | 15.0000  | 4250761.0000 |
| quantity_log                                                                     | 2670  | 7.4221      | 2.4557      | 2.7081   | 15.2626      |
| revenue (1000 R\$)                                                               | 2670  | 6508.4120   | 25829.3064  | 2.0000   | 267798.0000  |
| revenue_log                                                                      | 2670  | 5.0993      | 2.3588      | 0.6931   | 12.4980      |
| <i>Panel C. Municipality-by-year crop production data: Cassava (2005-2019)</i>   |       |             |             |          |              |
| area_planted (hectares)                                                          | 10335 | 904.7941    | 2083.6334   | 3.0000   | 45000.0000   |
| area_planted_log                                                                 | 10335 | 5.6360      | 1.5542      | 1.0986   | 10.7144      |
| area_harvested (hectares)                                                        | 10335 | 888.8364    | 2063.7582   | 3.0000   | 45000.0000   |
| area_harvested_log                                                               | 10335 | 5.6136      | 1.5548      | 1.0986   | 10.7144      |
| quantity (metric tons)                                                           | 10335 | 12373.8988  | 31966.3296  | 21.0000  | 900000.0000  |
| quantity_log                                                                     | 10335 | 8.1563      | 1.5884      | 3.0445   | 13.7101      |
| revenue (1000 R\$)                                                               | 10335 | 4380.8158   | 12083.2227  | 4.0000   | 317272.0000  |
| revenue_log                                                                      | 10335 | 7.0320      | 1.7165      | 1.3863   | 12.6675      |
| <i>Panel D. Municipality-by-year crop production data: Rice (2005-2019)</i>      |       |             |             |          |              |
| area_planted (hectares)                                                          | 7440  | 1594.8905   | 3446.3446   | 1.0000   | 91229.0000   |
| area_planted_log                                                                 | 7440  | 6.0941      | 1.8272      | 0.0000   | 11.4211      |
| area_harvested (hectares)                                                        | 7440  | 1572.2215   | 3279.0199   | 1.0000   | 53105.0000   |
| area_harvested_log                                                               | 7440  | 6.0816      | 1.8290      | 0.0000   | 10.8800      |
| quantity (metric tons)                                                           | 7440  | 3841.0032   | 13207.1700  | 1.0000   | 366846.0000  |
| quantity_log                                                                     | 7440  | 6.5340      | 2.0549      | 0.0000   | 12.8127      |
| revenue (1000 R\$)                                                               | 7440  | 2249.2070   | 9253.2282   | 1.0000   | 346389.0000  |
| revenue_log                                                                      | 7440  | 6.0061      | 2.0054      | 0.0000   | 12.7553      |

**Table A3—Summary Stats of Other Outcomes**

|                                                                                    | N     | Mean          | SD            | Min    | Max           |
|------------------------------------------------------------------------------------|-------|---------------|---------------|--------|---------------|
| <i>Panel A. Municipality-by-year cattle production data (2005-2019)</i>            |       |               |               |        |               |
| head_count                                                                         | 11505 | 103593.5      | 157921.1      | 10     | 2282445       |
| head_count_log                                                                     | 11505 | 10.5203       | 1.7201        | 2.3026 | 14.6408       |
| <i>Panel B. Municipality-by-year rural credit data (2005-2011 &amp; 2014-2019)</i> |       |               |               |        |               |
| credit_amounts_crop (R\$)                                                          | 9789  | 11339719.7656 | 46943701.1100 | 0.0000 | 1.1618e+09    |
| credit_amounts_crop_asinh                                                          | 9789  | 13.0315       | 4.8839        | 0.0000 | 21.5663       |
| credit_contracts_crop                                                              | 9789  | 237.3666      | 469.4125      | 0      | 9071          |
| credit_contracts_crop_asinh                                                        | 9789  | 4.7131        | 2.1367        | 0.0000 | 9.8060        |
| credit_amounts_cattle (R\$)                                                        | 9789  | 8855534.6892  | 16868245.1635 | 0.0000 | 5.3613e+08    |
| credit_amounts_cattle_asinh                                                        | 9789  | 14.5388       | 4.1214        | 0.0000 | 20.7930       |
| credit_contracts_cattle                                                            | 9789  | 303.4442      | 478.9730      | 0      | 9071          |
| credit_contracts_cattle_asinh                                                      | 9789  | 5.3844        | 1.8705        | 0.0000 | 9.8060        |
| amounts_pronaf (R\$, 2014-2019)                                                    | 4614  | 3493982.5780  | 6317141.8148  | 0.0000 | 94089088.0000 |
| amounts_pronaf_asinh (2014-2019)                                                   | 4614  | 13.8118       | 3.9065        | 0.0000 | 19.0529       |
| contracts_pronaf (2014-2019)                                                       | 4614  | 188.5847      | 260.4328      | 0      | 2951          |
| contracts_pronaf_asinh (2014-2019)                                                 | 4614  | 4.8065        | 1.9333        | 0.0000 | 8.6830        |
| amounts_proirriga (R\$, 2014-2019)                                                 | 4614  | 60677.3661    | 602921.8377   | 0.0000 | 17011092.0000 |
| amounts_proirriga_asinh (2014-2019)                                                | 4614  | 0.4205        | 2.4424        | 0.0000 | 17.3425       |
| contracts_proirriga (2014-2019)                                                    | 4614  | 0.0557        | 0.4244        | 0      | 10            |
| contracts_proirriga_asinh (2014-2019)                                              | 4614  | 0.0362        | 0.2286        | 0.0000 | 2.9982        |
| <i>Panel C. Municipality-by-year irrigation data (2005-2019)</i>                   |       |               |               |        |               |
| alltemp_irrigated_areashare                                                        | 11550 | 0.0876        | 0.7495        | 0.0000 | 27.3691       |
| soybean_irrigated_areashare                                                        | 11550 | 0.0280        | 0.2291        | 0.0000 | 7.7409        |
| rice_irrigated_areashare                                                           | 11550 | 0.0340        | 0.6801        | 0.0000 | 27.3491       |
| sugarcane_irrigated_areashare                                                      | 11550 | 0.0007        | 0.0299        | 0.0000 | 1.8775        |
| othertemp_irrigated_areashare                                                      | 11550 | 0.0249        | 0.1843        | 0.0000 | 5.5353        |

**Table A4 — Summary Stats of Control Variables**

|                                                                         | N     | Mean    | SD      | Min     | Max      |
|-------------------------------------------------------------------------|-------|---------|---------|---------|----------|
| <i>Panel A. Municipality-by-year policies (2005-2019)</i>               |       |         |         |         |          |
| protected_areashare                                                     | 11565 | 0.2258  | 0.3092  | 0.0000  | 1.0000   |
| green (dummy)                                                           | 11565 | 0.0874  | 0.2825  | 0.0000  | 1.0000   |
| blacklist (dummy)                                                       | 11565 | 0.0353  | 0.1845  | 0.0000  | 1.0000   |
| creditsoy2008 (dummy)                                                   | 11565 | 0.5181  | 0.4834  | 0.0000  | 1.0000   |
| <i>Panel B. Municipality-by-year crop and cattle prices (2005-2019)</i> |       |         |         |         |          |
| sem1_rprice_index_crop                                                  | 11565 | -0.0351 | 1.2593  | -0.8363 | 18.9797  |
| sem1_rprice_index_crop_lag                                              | 10794 | -0.0269 | 1.2703  | -0.8349 | 19.0799  |
| sem2_rprice_index_crop                                                  | 11565 | -0.0393 | 1.2544  | -0.8319 | 18.9989  |
| sem2_rprice_index_crop_lag                                              | 10794 | -0.0349 | 1.2579  | -0.8323 | 18.8887  |
| sem1_rprice_index_cattle                                                | 11565 | 40.2700 | 45.1783 | 0.0000  | 345.7506 |
| sem1_rprice_index_cattle_lag                                            | 10794 | 40.0586 | 45.0035 | 0.0000  | 345.7506 |
| sem2_rprice_index_cattle                                                | 11565 | 41.3486 | 46.1448 | 0.0000  | 335.3997 |
| sem2_rprice_index_cattle_lag                                            | 10794 | 41.0357 | 45.8250 | 0.0000  | 335.3997 |



## B — Main Results Tables

**Table B1 — Impacts of Year-Round RZSM Deficits on Deforestation<sup>a</sup>**

|                                                      | average effects       | heterogeneous effects   |
|------------------------------------------------------|-----------------------|-------------------------|
| all Legal Amazon municipalities                      | 0.00014*<br>(0.00008) |                         |
| municipalities with more family farms <sup>b</sup>   |                       | 0.00028***<br>(0.00009) |
| municipalities with less family farms                |                       | -0.00005<br>(0.00009)   |
| difference between the two                           |                       | 0.00032***<br>(0.00012) |
| one lag of the outcome variable                      | YES                   | YES                     |
| one lag of forest stock ( <i>asinh</i> )             | YES                   | YES                     |
| price controls <sup>c</sup>                          | YES                   | YES                     |
| policy controls <sup>d</sup>                         | YES                   | YES                     |
| year fixed effects                                   | YES                   | YES                     |
| municipality-by-RZSM-grid fixed effects <sup>e</sup> | YES                   | YES                     |
| <i>adj.R</i> <sup>2</sup>                            | 0.89                  | 0.89                    |
| within <i>R</i> <sup>2</sup>                         | 0.29                  | 0.29                    |
| <i>N</i>                                             | 184,240               | 184,030 <sup>f</sup>    |

*Note:* <sup>a</sup>We construct our forest losses measure based on the annual land cover data obtained from MapBiomas. We focus on forest loss relative to the base year of 2000 from 2005 to 2019 in the Legal Amazon. We take *asinh* to deal with zero values. <sup>b</sup>We classify a municipality as one with more family farms if its family farm area share is above the median. We calculate family farm areas and their shares within each municipality based on the 2006 Brazilian Agriculture Census data. <sup>c</sup>We control for local exposures to exogenous crop and cattle prices following Assunção et al. (2015, 2020). <sup>d</sup>We control for major municipality, regional, national and international regulation policies, including blacklisting and green municipality programs, the 2008 credit concession inside the Amazon Biome, and national and international protected area recognitions. For the first two, we use their lagged implementation statuses. For protected area, we use the lag of the area share as the control. <sup>e</sup>In total, we have 13,160 municipality-by-RZSM-grid cells and 771 municipalities. <sup>f</sup>One Legal Amazon municipality does not show up in the 2006 Brazilian Agriculture Census data. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Table B2 — Impacts of Year-Round RZSM Deficits on Forest-Related Environmental Fines<sup>a</sup>**

|                                          | fine counts           |                       | fine amounts          |                       |
|------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                          | average effects       | heterogeneous effects | average effects       | heterogeneous effects |
| all Legal Amazon municipalities          | 0.0034***<br>(0.0012) |                       | 0.0185***<br>(0.0067) |                       |
| municipalities with more family farms    |                       | 0.0044***<br>(0.0016) |                       | 0.0222**<br>(0.0090)  |
| municipalities with less family farms    |                       | 0.0017<br>(0.0014)    |                       | 0.0120<br>(0.077)     |
| difference between the two               |                       | 0.0028<br>(0.0021)    |                       | 0.0101<br>(0.0114)    |
| one lag of the outcome variable          | YES                   | YES                   | YES                   | YES                   |
| one lag of forest stock ( <i>asinh</i> ) | YES                   | YES                   | YES                   | YES                   |
| price controls                           | YES                   | YES                   | YES                   | YES                   |
| policy controls                          | YES                   | YES                   | YES                   | YES                   |
| year fixed effects                       | YES                   | YES                   | YES                   | YES                   |
| municipality fixed effects               | YES                   | YES                   | YES                   | YES                   |
| <i>adj.R</i> <sup>2</sup>                | 0.69                  | 0.69                  | 0.56                  | 0.56                  |
| within <i>R</i> <sup>2</sup>             | 0.03                  | 0.03                  | 0.01                  | 0.01                  |
| <i>N</i>                                 | 10,794                | 10,780 <sup>b</sup>   | 10,794                | 10,780                |

*Note:* <sup>a</sup>We obtain forest-related environmental fines from the Brazilian Institute of the Environment and Renewable Natural Resources. We take *asinh* to deal with zero values. <sup>b</sup>One Legal Amazon municipality does not show up in the 2006 Brazilian Agriculture Census data. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table B3 — Impacts of Seasonal RZSM Deficits on Deforestation and Forest-Related Environmental Fines<sup>a</sup>

|                                        | deforestation |                 | fines amount |                 | fines count  |                 |
|----------------------------------------|---------------|-----------------|--------------|-----------------|--------------|-----------------|
|                                        | avg. effects  | hetero. effects | avg. effects | hetero. effects | avg. effects | hetero. effects |
| <i>Pregrowing-season RZSM deficits</i> |               |                 |              |                 |              |                 |
| all Legal Amazon                       | -0.00079**    |                 | 0.1249       |                 | 0.0557       |                 |
| municipalities                         | (0.00035)     |                 | (0.5315)     |                 | (0.1005)     |                 |
| municipalities with                    |               | -0.00058***     |              | -1.4477***      |              | -0.1868***      |
| more family farms                      |               | (0.00020)       |              | (0.2085)        |              | (0.0372)        |
| municipalities with                    |               | -0.00124*       |              | 0.6420          |              | 0.1353          |
| less family farms                      |               | (0.00068)       |              | (0.4532)        |              | (0.0967)        |
| difference between the two             |               | 0.00066         |              | -2.0897***      |              | -0.3221***      |
|                                        |               | (0.00068)       |              | (0.4989)        |              | (0.1037)        |
| <i>Growing-season RZSM deficits</i>    |               |                 |              |                 |              |                 |
| all Legal Amazon                       | 0.00014*      |                 | 0.0184***    |                 | 0.0034***    |                 |
| municipalities                         | (0.00008)     |                 | (0.0067)     |                 | (0.0012)     |                 |
| municipalities with                    |               | 0.00028***      |              | 0.0230***       |              | 0.0046***       |
| more family farms                      |               | (0.00009)       |              | (0.0088)        |              | (0.0016)        |
| municipalities with                    |               | -0.00005        |              | 0.0117          |              | 0.0016          |
| less family farms                      |               | (0.00009)       |              | (0.0077)        |              | (0.0014)        |
| difference between the two             |               | 0.00032***      |              | 0.0113          |              | 0.0030          |
|                                        |               | (0.00012)       |              | (0.0113)        |              | (0.0020)        |
| lagged outcome                         | YES           | YES             | YES          | YES             | YES          | YES             |
| controls                               | YES           | YES             | YES          | YES             | YES          | YES             |
| fixed effects                          | YES           | YES             | YES          | YES             | YES          | YES             |
| <i>adj.R</i> <sup>2</sup>              | 0.89          | 0.89            | 0.56         | 0.56            | 0.69         | 0.69            |
| within <i>R</i> <sup>2</sup>           | 0.29          | 0.29            | 0.01         | 0.01            | 0.03         | 0.03            |
| <i>N</i>                               | 184,240       | 184,030         | 10,794       | 10,780          | 10,794       | 10,780          |

*Note:* <sup>a</sup>We use the same deforestation and environmental fine measures as in Table B1 and B2. But we replace year-round RZSM deficits with two seasonal RZSM deficits, one for the pregrowing season (July-August) and one for the growing season (all other months in the same year). Regressions include municipality and year fixed effects, lagged dependent variable, and various controls as the other tables above. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

# C — Mechanism Results Tables

**Table C1 — Impacts of Seasonal RZSM Deficits on Rice Production<sup>a</sup>**

|                                        | area planted        |                        | area harvested      |                        | output quantity     |                        | output revenue      |                        | average price        |                        |
|----------------------------------------|---------------------|------------------------|---------------------|------------------------|---------------------|------------------------|---------------------|------------------------|----------------------|------------------------|
|                                        | average effects     | heterogeneous effects  | average effects     | heterogeneous effects  | average effects     | heterogeneous effects  | average effects     | heterogeneous effects  | average effects      | heterogeneous effects  |
| <i>Pregrowing-season RZSM deficits</i> |                     |                        |                     |                        |                     |                        |                     |                        |                      |                        |
| all Legal Amazon municipalities        | -0.2162<br>(0.1709) |                        | -0.2328<br>(0.1711) |                        | -0.0803<br>(0.1429) |                        | -0.0796<br>(0.1482) |                        | -0.0050<br>(0.0180)  |                        |
| municipalities with more family farms  |                     | -0.8957**<br>(0.3702)  |                     | -0.9080**<br>(0.3727)  |                     | -1.0897***<br>(0.2593) |                     | -1.1162***<br>(0.3204) |                      | -0.0944***<br>(0.0128) |
| municipalities with less family farms  |                     | -0.1318<br>(0.1556)    |                     | -0.1487<br>(0.1555)    |                     | -0.0214<br>(0.1233)    |                     | -0.0144<br>(0.1288)    |                      | 0.0001<br>(0.0183)     |
| difference between the two             |                     | -0.7639*<br>(0.4002)   |                     | -0.7593*<br>(0.4025)   |                     | -1.0683***<br>(0.2851) |                     | -1.1018***<br>(0.3443) |                      | -0.0946***<br>(0.0176) |
| <i>Growing-season RZSM deficits</i>    |                     |                        |                     |                        |                     |                        |                     |                        |                      |                        |
| all Legal Amazon municipalities        | -0.0010<br>(0.0032) |                        | -0.0003<br>(0.0032) |                        | 0.0025<br>(0.0035)  |                        | 0.0040<br>(0.0031)  |                        | 0.0011**<br>(0.0005) |                        |
| municipalities with more family farms  |                     | -0.0105**<br>(0.0042)  |                     | -0.0100**<br>(0.0042)  |                     | -0.0094**<br>(0.0036)  |                     | -0.0067<br>(0.0052)    |                      | -0.0004<br>(0.0004)    |
| municipalities with less family farms  |                     | 0.0045**<br>(0.0019)   |                     | 0.0053**<br>(0.0021)   |                     | 0.0071***<br>(0.0027)  |                     | 0.0076***<br>(0.0028)  |                      | 0.0016***<br>(0.0005)  |
| difference between the two             |                     | -0.0150***<br>(0.0045) |                     | -0.0152***<br>(0.0046) |                     | -0.0165***<br>(0.0042) |                     | -0.0142**<br>(0.0057)  |                      | -0.0021***<br>(0.0006) |
| lagged outcome                         | YES                 | YES                    | YES                 | YES                    | YES                 | YES                    | YES                 | YES                    | YES                  | YES                    |
| controls                               | YES                 | YES                    | YES                 | YES                    | YES                 | YES                    | YES                 | YES                    | YES                  | YES                    |
| fixed effects                          | YES                 | YES                    | YES                 | YES                    | YES                 | YES                    | YES                 | YES                    | YES                  | YES                    |
| adj. R <sup>2</sup>                    | 0.85                | 0.85                   | 0.85                | 0.85                   | 0.86                | 0.86                   | 0.85                | 0.85                   | 0.76                 | 0.76                   |
| within R <sup>2</sup>                  | 0.43                | 0.43                   | 0.42                | 0.42                   | 0.40                | 0.40                   | 0.40                | 0.40                   | 0.10                 | 0.11                   |
| N <sup>b</sup>                         | 6,944               | 6,944                  | 6,944               | 6,944                  | 6,944               | 6,944                  | 6,944               | 6,944                  | 6,944                | 6,944                  |

Note: <sup>a</sup>We obtain annual crop production data from the Brazilian Institute of Geography and Statistics and take log for all production measures. Each regression is weighted by the 2005 municipality-specific outcome level. For price regressions, we use 2005 municipality-specific output weights.

<sup>b</sup>There are 496 Legal Amazon municipalities that have consistently-reported rice production data. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table C2 — Impacts of Year-Round RZSM Deficits on Cassava Production<sup>a</sup>

|                                       | area planted    |                       | area harvested  |                       | output quantity |                       | output revenue  |                       | average price   |                       |
|---------------------------------------|-----------------|-----------------------|-----------------|-----------------------|-----------------|-----------------------|-----------------|-----------------------|-----------------|-----------------------|
|                                       | average effects | heterogeneous effects | average effects | heterogeneous effects | average effects | heterogeneous effects | average effects | heterogeneous effects | average effects | heterogeneous effects |
| all Legal Amazon municipalities       | -0.0028*        |                       | -0.0029*        |                       | -0.0034**       |                       | -0.0016         |                       | 0.0021***       |                       |
|                                       | (0.0016)        |                       | (0.0016)        |                       | (0.0015)        |                       | (0.0011)        |                       | (0.0008)        |                       |
| municipalities with more family farms |                 | -0.0030               |                 | -0.0030               |                 | -0.0037*              |                 | -0.0020               |                 | 0.0019*               |
|                                       |                 | (0.0019)              |                 | (0.0020)              |                 | (0.0020)              |                 | (0.0014)              |                 | (0.0011)              |
| municipalities with less family farms |                 | -0.0022               |                 | -0.0025               |                 | -0.0028*              |                 | -0.0003               |                 | 0.0026***             |
|                                       |                 | (0.0016)              |                 | (0.0016)              |                 | (0.0017)              |                 | (0.0015)              |                 | (0.0007)              |
| difference between the two            |                 | -0.0008               |                 | -0.0005               |                 | -0.0009               |                 | -0.0017               |                 | -0.0007               |
|                                       |                 | (0.0024)              |                 | (0.0024)              |                 | (0.0024)              |                 | (0.0020)              |                 | (0.0012)              |
| lagged outcome                        | YES             | YES                   | YES             | YES                   | YES             | YES                   | YES             | YES                   | YES             | YES                   |
| controls                              | YES             | YES                   | YES             | YES                   | YES             | YES                   | YES             | YES                   | YES             | YES                   |
| fixed effects                         | YES             | YES                   | YES             | YES                   | YES             | YES                   | YES             | YES                   | YES             | YES                   |
| adj. $R^2$                            | 0.93            | 0.93                  | 0.92            | 0.92                  | 0.92            | 0.92                  | 0.86            | 0.86                  | 0.79            | 0.79                  |
| within $R^2$                          | 0.47            | 0.47                  | 0.42            | 0.42                  | 0.36            | 0.36                  | 0.35            | 0.35                  | 0.20            | 0.20                  |
| $N^b$                                 | 9,646           | 9,646                 | 9,646           | 9,646                 | 9,646           | 9,646                 | 9,646           | 9,646                 | 9,646           | 9,646                 |

*Note:* <sup>a</sup>We obtain annual crop production data from the Brazilian Institute of Geography and Statistics and take log for all production measures. RZSM deficit intensity is measured over the full calendar year rather than separate pregrowing/growing windows, reflecting cassava's continuous, multi-year production cycle. As before, each regression is weighted by the 2005 municipality-specific outcome level. For price regressions, we use 2005 municipality-specific output weights. <sup>b</sup>There are 689 Legal Amazon municipalities that have consistently-reported cassava production data. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table C3 — Impacts of Seasonal RZSM Deficits on Soybean Production<sup>a</sup>

|                                        | area planted       |                        | area harvested     |                        | output quantity    |                        | output revenue       |                        | average price         |                        |
|----------------------------------------|--------------------|------------------------|--------------------|------------------------|--------------------|------------------------|----------------------|------------------------|-----------------------|------------------------|
|                                        | average effects    | heterogeneous effects  | average effects    | heterogeneous effects  | average effects    | heterogeneous effects  | average effects      | heterogeneous effects  | average effects       | heterogeneous effects  |
| <i>Pregrowing-season RZSM deficits</i> |                    |                        |                    |                        |                    |                        |                      |                        |                       |                        |
| all Legal Amazon municipalities        | 0.0813<br>(0.0776) |                        | 0.0846<br>(0.0774) |                        | 0.1327<br>(0.1239) |                        | 0.2098<br>(0.1422)   |                        | 0.0727***<br>(0.0276) |                        |
| municipalities with more family farms  |                    | -0.1373***<br>(0.0082) |                    | -0.1368***<br>(0.0084) |                    | -0.0840***<br>(0.0141) |                      | -0.0038<br>(0.0185)    |                       | 0.0544***<br>(0.0085)  |
| municipalities with less family farms  |                    | 0.1522***<br>(0.0387)  |                    | 0.1564***<br>(0.0385)  |                    | 0.1990*<br>(0.1120)    |                      | 0.2819**<br>(0.1350)   |                       | 0.0783**<br>(0.0327)   |
| difference between the two             |                    | -0.2895***<br>(0.0370) |                    | -0.2932***<br>(0.0369) |                    | -0.2831**<br>(0.1106)  |                      | -0.2857**<br>(0.1319)  |                       | -0.0239<br>(0.0320)    |
| <i>Growing-season RZSM deficits</i>    |                    |                        |                    |                        |                    |                        |                      |                        |                       |                        |
| all Legal Amazon municipalities        | 0.0009<br>(0.0021) |                        | 0.0008<br>(0.0022) |                        | 0.0031<br>(0.0025) |                        | 0.0067**<br>(0.0032) |                        | 0.0008<br>(0.0009)    |                        |
| municipalities with more family farms  |                    | -0.0747***<br>(0.0053) |                    | -0.0747***<br>(0.0053) |                    | -0.0688***<br>(0.0069) |                      | -0.2581***<br>(0.0114) |                       | -0.1727***<br>(0.0054) |
| municipalities with less family farms  |                    | 0.0009<br>(0.0021)     |                    | 0.0008<br>(0.0022)     |                    | 0.0031<br>(0.0025)     |                      | 0.0067**<br>(0.0032)   |                       | 0.0008<br>(0.0009)     |
| difference between the two             |                    | -0.0755***<br>(0.0058) |                    | -0.0754***<br>(0.0058) |                    | -0.0719***<br>(0.0077) |                      | -0.2648***<br>(0.0123) |                       | -0.1735***<br>(0.0056) |
| lagged outcome                         | YES                | YES                    | YES                | YES                    | YES                | YES                    | YES                  | YES                    | YES                   | YES                    |
| controls                               | YES                | YES                    | YES                | YES                    | YES                | YES                    | YES                  | YES                    | YES                   | YES                    |
| fixed effects                          | YES                | YES                    | YES                | YES                    | YES                | YES                    | YES                  | YES                    | YES                   | YES                    |
| adj. R <sup>2</sup>                    | 0.98               | 0.98                   | 0.98               | 0.98                   | 0.98               | 0.98                   | 0.97                 | 0.97                   | 0.95                  | 0.95                   |
| within R <sup>2</sup>                  | 0.68               | 0.68                   | 0.68               | 0.68                   | 0.62               | 0.62                   | 0.52                 | 0.52                   | 0.05                  | 0.05                   |
| N <sup>b</sup>                         | 2,338              | 2,338                  | 2,338              | 2,338                  | 2,338              | 2,338                  | 2,338                | 2,338                  | 2,338                 | 2,338                  |

Note: <sup>a</sup>We obtain annual crop production data from the Brazilian Institute of Geography and Statistics and take log for all production measures. Each regression is weighted by the 2005 municipality-specific outcome level. <sup>b</sup>There are 167 Legal Amazon municipalities that have consistently-reported soybean production data. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

Table C4 — Impacts of Year-Round RZSM Deficits on Sugarcane Production<sup>a</sup>

|                                       | area planted        |                       | area harvested      |                       | output quantity    |                       | output revenue     |                       | average price       |                        |
|---------------------------------------|---------------------|-----------------------|---------------------|-----------------------|--------------------|-----------------------|--------------------|-----------------------|---------------------|------------------------|
|                                       | average effects     | heterogeneous effects | average effects     | heterogeneous effects | average effects    | heterogeneous effects | average effects    | heterogeneous effects | average effects     | heterogeneous effects  |
| all Legal Amazon municipalities       | -0.0001<br>(0.0006) |                       | -0.0002<br>(0.0007) |                       | 0.0004<br>(0.0007) |                       | 0.0004<br>(0.0005) |                       | -0.0004<br>(0.0005) |                        |
| municipalities with more family farms |                     | -0.0001<br>(0.0006)   |                     | -0.0002<br>(0.0006)   |                    | 0.0003<br>(0.0006)    |                    | 0.0004<br>(0.0005)    |                     | -0.0001<br>(0.0003)    |
| municipalities with less family farms |                     | -0.0002<br>(0.0011)   |                     | -0.0004<br>(0.0012)   |                    | 0.0009<br>(0.0013)    |                    | -0.0028*<br>(0.0015)  |                     | -0.0029***<br>(0.0008) |
| difference between the two            |                     | 0.0001<br>(0.0008)    |                     | 0.0002<br>(0.0009)    |                    | -0.0005<br>(0.0010)   |                    | 0.0032**<br>(0.0013)  |                     | 0.0028***<br>(0.0008)  |
| lagged outcome                        | YES                 | YES                   | YES                 | YES                   | YES                | YES                   | YES                | YES                   | YES                 | YES                    |
| controls                              | YES                 | YES                   | YES                 | YES                   | YES                | YES                   | YES                | YES                   | YES                 | YES                    |
| fixed effects                         | YES                 | YES                   | YES                 | YES                   | YES                | YES                   | YES                | YES                   | YES                 | YES                    |
| <i>adj. R</i> <sup>2</sup>            | 0.96                | 0.96                  | 0.96                | 0.96                  | 0.95               | 0.95                  | 0.94               | 0.94                  | 0.86                | 0.86                   |
| within <i>R</i> <sup>2</sup>          | 0.51                | 0.51                  | 0.47                | 0.47                  | 0.50               | 0.50                  | 0.55               | 0.55                  | 0.42                | 0.42                   |
| <i>N</i> <sup>b</sup>                 | 2,492               | 2,492                 | 2,492               | 2,492                 | 2,492              | 2,492                 | 2,492              | 2,492                 | 2,492               | 2,492                  |

*Note:* <sup>a</sup>We obtain annual crop production data from the Brazilian Institute of Geography and Statistics and take log for all production measures. RZSM deficit intensity is measured over the full calendar year rather than separate pregrowing/growing windows, reflecting sugarcane's continuous, multi-year production cycle. As before, each regression is weighted by the 2005 municipality-specific outcome level. For price regressions, we use 2005 municipality-specific output weights. <sup>b</sup>There are 178 Legal Amazon municipalities that have consistently-reported sugarcane production data. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Table C5 — Impacts of Year-Round RZSM Deficits on Cattle Production<sup>a</sup>**

|                                          | average effects          | heterogeneous effects    |
|------------------------------------------|--------------------------|--------------------------|
| all Legal Amazon municipalities          | -0.00058***<br>(0.00016) |                          |
| municipalities with more family farms    |                          | -0.00022<br>(0.00031)    |
| municipalities with less family farms    |                          | -0.00070***<br>(0.00018) |
| difference between the two               |                          | 0.00047<br>(0.00036)     |
| one lag of the outcome variable          | YES                      | YES                      |
| one lag of forest stock ( <i>asinh</i> ) | YES                      | YES                      |
| price controls                           | YES                      | YES                      |
| policy controls                          | YES                      | YES                      |
| year fixed effects                       | YES                      | YES                      |
| municipality fixed effects <sup>e</sup>  | YES                      | YES                      |
| <i>adj.R</i> <sup>2</sup>                | 0.98                     | 0.98                     |
| within <i>R</i> <sup>2</sup>             | 0.30                     | 0.30                     |
| <i>N</i>                                 | 10,738                   | 10,738                   |

*Note:* <sup>a</sup>We obtain annual cattle head count data from the Brazilian Institute of Geography and Statistics. We focus on 767 Legal Amazon municipalities that have consistently reported positive cattle production across years and take log of cattle head counts. Our regression is weighted by the 2005 municipality-specific cattle head count. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .



Table C6 — Impacts of RZSM Deficits on Cassava Production in Rice-Cassava Municipalities<sup>a</sup>

|                                                    | area planted         |                        | area harvested        |                       | output quantity       |                       | output revenue       |                       | average price      |                       |
|----------------------------------------------------|----------------------|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|----------------------|-----------------------|--------------------|-----------------------|
|                                                    | average effects      | heterogeneous effects  | average effects       | heterogeneous effects | average effects       | heterogeneous effects | average effects      | heterogeneous effects | average effects    | heterogeneous effects |
| <i>Pregrowing-season RZSM deficits<sup>b</sup></i> |                      |                        |                       |                       |                       |                       |                      |                       |                    |                       |
| all Legal Amazon municipalities                    | 0.0347**<br>(0.0145) |                        | 0.0617***<br>(0.0133) |                       | 0.0460***<br>(0.0116) |                       | 0.0399**<br>(0.0196) |                       | 0.0098<br>(0.0186) |                       |
| municipalities with more family farms              |                      | 0.0033<br>(0.0088)     |                       | 0.0437***<br>(0.0114) |                       | 0.0341**<br>(0.0159)  |                      | 0.0173<br>(0.0152)    |                    | -0.0146<br>(0.0146)   |
| municipalities with less family farms              |                      | 0.0493***<br>(0.0126)  |                       | 0.0695***<br>(0.0164) |                       | 0.0492***<br>(0.0156) |                      | 0.0648***<br>(0.0230) |                    | 0.0183<br>(0.0229)    |
| difference between the two                         |                      | -0.0460***<br>(0.0149) |                       | -0.0258<br>(0.0197)   |                       | -0.0151<br>(0.0228)   |                      | -0.0475*<br>(0.0243)  |                    | -0.0329<br>(0.0250)   |
| year-round RZSM deficits                           | YES                  | YES                    | YES                   | YES                   | YES                   | YES                   | YES                  | YES                   | YES                | YES                   |
| lagged outcome                                     | YES                  | YES                    | YES                   | YES                   | YES                   | YES                   | YES                  | YES                   | YES                | YES                   |
| controls                                           | YES                  | YES                    | YES                   | YES                   | YES                   | YES                   | YES                  | YES                   | YES                | YES                   |
| fixed effects                                      | YES                  | YES                    | YES                   | YES                   | YES                   | YES                   | YES                  | YES                   | YES                | YES                   |
| <i>adj. R</i> <sup>2</sup>                         | 0.94                 | 0.94                   | 0.94                  | 0.94                  | 0.93                  | 0.93                  | 0.88                 | 0.88                  | 0.81               | 0.81                  |
| within <i>R</i> <sup>2</sup>                       | 0.49                 | 0.49                   | 0.47                  | 0.47                  | 0.42                  | 0.42                  | 0.35                 | 0.35                  | 0.21               | 0.21                  |
| <i>N</i> <sup>b</sup>                              | 6,496                | 6,496                  | 6,496                 | 6,496                 | 6,496                 | 6,496                 | 6,496                | 6,496                 | 6,496              | 6,496                 |

Note: <sup>a</sup>We obtain annual crop production data from the Brazilian Institute of Geography and Statistics and take log for all production measures. Sample restricted to municipality-years where cassava and rice are jointly cultivated. <sup>b</sup>We define the pregrowing season here as July-August of the current year to see the contemporaneous response of cassava production to pregrowing-season drought shock. <sup>c</sup>There are 464 Legal Amazon municipalities that have consistently-reported cassava production data and rice-cassava intercropping. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Table C7 — Impacts of RZSM Deficits on Irrigated Area Share by Crop<sup>a</sup>**

|                                                    | soybean                 |                           | rice                    |                         | other temporary crops    |                          | sugarcane               |                         |
|----------------------------------------------------|-------------------------|---------------------------|-------------------------|-------------------------|--------------------------|--------------------------|-------------------------|-------------------------|
|                                                    | average effects         | heterogeneous effects     | average effects         | heterogeneous effects   | average effects          | heterogeneous effects    | average effects         | heterogeneous effects   |
| <i>Pregrowing-season RZSM deficits<sup>b</sup></i> |                         |                           |                         |                         |                          |                          |                         |                         |
| all Legal Amazon municipalities                    | 1.52e-04<br>(6.25e-04)  |                           | 2.59e-03<br>(3.01e-03)  |                         | -8.21e-04<br>(8.93e-04)  |                          |                         |                         |
| municipalities with more family farms              |                         | -8.49e-04**<br>(3.27e-04) |                         | -5.22e-04<br>(4.85e-04) |                          | -1.05e-03<br>(1.16e-03)  |                         |                         |
| municipalities with less family farms              |                         | 4.84e-04<br>(8.34e-04)    |                         | 3.62e-03<br>(4.09e-03)  |                          | -7.42e-04<br>(1.07e-03)  |                         |                         |
| difference between the two                         |                         | -1.33e-03<br>(8.66e-04)   |                         | -4.14e-03<br>(4.11e-03) |                          | -3.09e-04<br>(1.43e-03)  |                         |                         |
| <i>Growing-season RZSM deficits<sup>b</sup></i>    |                         |                           |                         |                         |                          |                          |                         |                         |
| all Legal Amazon municipalities                    | -6.07e-06<br>(1.26e-05) |                           | -3.22e-05<br>(2.40e-05) |                         | -2.79e-05*<br>(1.53e-05) |                          |                         |                         |
| municipalities with more family farms              |                         | -1.03e-05<br>(1.25e-05)   |                         | -3.28e-05<br>(2.52e-05) |                          | -3.22e-05*<br>(1.91e-05) |                         |                         |
| municipalities with less family farms              |                         | 2.31e-06<br>(1.36e-05)    |                         | -2.86e-05<br>(2.19e-05) |                          | -1.98e-05<br>(1.46e-05)  |                         |                         |
| difference between the two                         |                         | -1.26e-05*<br>(7.20e-06)  |                         | -4.21e-06<br>(6.80e-06) |                          | -1.24e-05<br>(1.82e-05)  |                         |                         |
| <i>Year-round RZSM deficits</i>                    |                         |                           |                         |                         |                          |                          |                         |                         |
| all Legal Amazon municipalities                    |                         |                           |                         |                         |                          |                          | -3.08e-06<br>(3.13e-06) |                         |
| municipalities with more family farms              |                         |                           |                         |                         |                          |                          |                         | -1.47e-06<br>(2.19e-06) |
| municipalities with less family farms              |                         |                           |                         |                         |                          |                          |                         | -5.95e-06<br>(7.11e-06) |
| difference between the two                         |                         |                           |                         |                         |                          |                          |                         | 4.48e-06<br>(6.98e-06)  |
| lagged outcome                                     | YES                     | YES                       | YES                     | YES                     | YES                      | YES                      | YES                     | YES                     |
| controls                                           | YES                     | YES                       | YES                     | YES                     | YES                      | YES                      | YES                     | YES                     |
| fixed effects                                      | YES                     | YES                       | YES                     | YES                     | YES                      | YES                      | YES                     | YES                     |
| <i>adj. R</i> <sup>2</sup>                         | 0.97                    | 0.97                      | 1.00                    | 1.00                    | 0.91                     | 0.91                     | 0.87                    | 0.87                    |
| within <i>R</i> <sup>2</sup>                       | 0.87                    | 0.87                      | 0.98                    | 0.98                    | 0.47                     | 0.47                     | 0.82                    | 0.82                    |
| <i>N</i> <sup>c</sup>                              | 10,780                  | 10,780                    | 10,780                  | 10,780                  | 10,780                   | 10,780                   | 10,780                  | 10,780                  |

*Note:* <sup>a</sup>The dependent variable is irrigated area share of total agricultural establishment area (2006 Agricultural Census), by crop. Regressions include municipality and year fixed effects, lagged dependent variable, and various controls as the other tables above. <sup>b</sup>To capture the contemporaneous responses of temporary crops' irrigation to agricultural drought, we use two seasonal RZSM deficits within the same year with the pregrowing season from July to August and the growing season covering all other months in the same year. <sup>c</sup>All 771 municipalities are matched with agricultural establishment data; the number of observations is 10,780. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Table C8 — Impacts of Year-Round RZSM Deficits on Rural Credit Usage<sup>a</sup>**

|                                       | crop                  |                        | cattle                |                        | PRONAF <sup>b</sup> |                       | Proirriga <sup>b</sup> |                     |
|---------------------------------------|-----------------------|------------------------|-----------------------|------------------------|---------------------|-----------------------|------------------------|---------------------|
|                                       | amounts               | contracts              | amounts               | contracts              | amounts             | contracts             | amounts                | contracts           |
| <i>Average effects</i>                |                       |                        |                       |                        |                     |                       |                        |                     |
| all Legal Amazon municipalities       | -0.0130**<br>(0.0063) | -0.0052***<br>(0.0015) | -0.0152**<br>(0.0071) | -0.0056***<br>(0.0015) | 0.0008<br>(0.0053)  | -0.0021<br>(0.0015)   | 0.0001<br>(0.0014)     | -0.0000<br>(0.0001) |
| <i>Heterogeneous effects</i>          |                       |                        |                       |                        |                     |                       |                        |                     |
| municipalities with more family farms | -0.0200**<br>(0.0077) | -0.0078***<br>(0.0022) | -0.0232**<br>(0.0102) | -0.0094***<br>(0.0018) | -0.0050<br>(0.0053) | -0.0038*<br>(0.0020)  | -0.0008<br>(0.0011)    | -0.0001<br>(0.0001) |
| municipalities with less family farms | -0.0006<br>(0.0105)   | -0.0007<br>(0.0015)    | -0.0010<br>(0.0040)   | 0.0013<br>(0.0014)     | 0.0110<br>(0.0100)  | 0.0009<br>(0.0015)    | 0.0018<br>(0.0034)     | 0.0001<br>(0.0003)  |
| difference                            | -0.0194<br>(0.0126)   | -0.0071***<br>(0.0026) | -0.0222**<br>(0.0107) | -0.0106***<br>(0.0022) | -0.0160<br>(0.0111) | -0.0048**<br>(0.0024) | -0.0027<br>(0.0034)    | -0.0002<br>(0.0003) |
| lagged outcome                        | YES                   | YES                    | YES                   | YES                    | YES                 | YES                   | YES                    | YES                 |
| controls                              | YES                   | YES                    | YES                   | YES                    | YES                 | YES                   | YES                    | YES                 |
| fixed effects                         | YES                   | YES                    | YES                   | YES                    | YES                 | YES                   | YES                    | YES                 |
| <i>N</i> <sup>c</sup>                 | 8,283                 | 8,283                  | 8,283                 | 8,283                  | 3,845               | 3,845                 | 3,845                  | 3,845               |

*Note:* <sup>a</sup>Credit amounts and number of contracts are obtained from the Central Bank of Brazil's Matriz de Dados do Crédito Rural (MDCR) for the years of 2014-2019, supplemented by the cattle and crop credit data of 2005-2011 from Assunção et al. (2020). To align with Assunção et al. (2020), I calculate annual credit measures for 2014-2019 based on the same year-round time window from last August to this July, which is almost aligned with our temporary-crop-year window from last July to this June. We take *asinh* to deal with zero values in credit amounts or number of contracts. Regressions include municipality and year fixed effects, lagged dependent variable, and various controls as the other tables above. <sup>b</sup>PRONAF is the main federal credit program for small-scale or family farmers and Proirriga is an investment credit line focused specifically on irrigation and climate-control infrastructure. <sup>c</sup>The number of municipality-year observations is higher for crop and cattle regressions as we only have data for PRONAF and Proirriga rural credit programs from 2014 to 2019. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

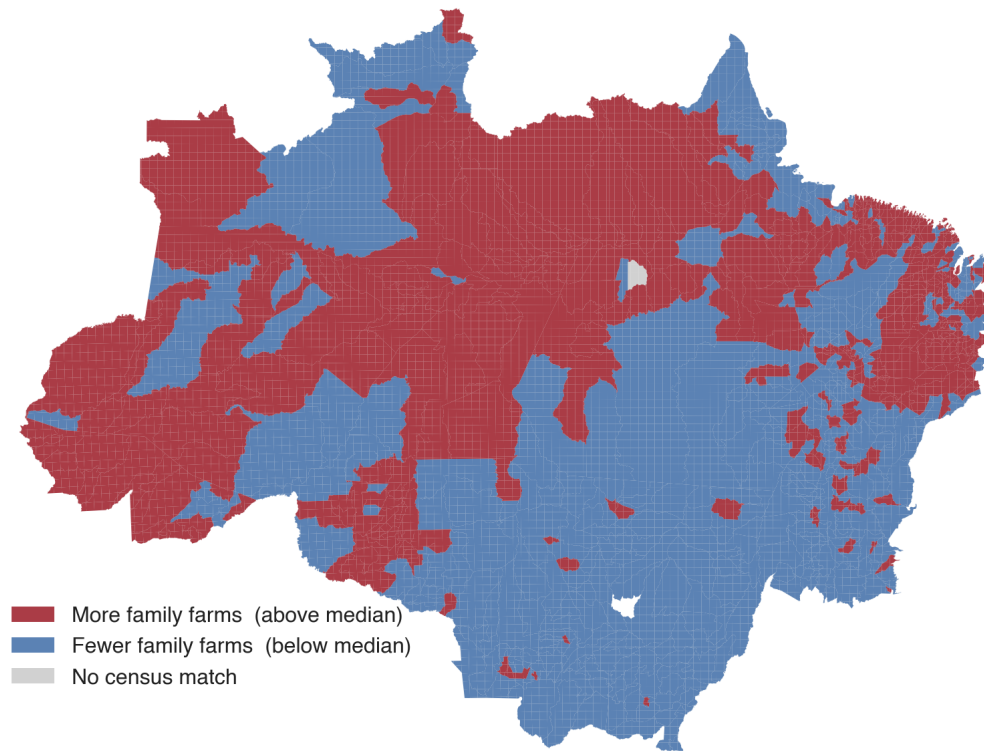
Table C9 — The Impacts of Seasonal RZSM Deficits on Rural Credit Usage<sup>a</sup>

|                                        | crop                  |                        | cattle                 |                        | PRONAF               |                        | Proirriga           |                      |
|----------------------------------------|-----------------------|------------------------|------------------------|------------------------|----------------------|------------------------|---------------------|----------------------|
|                                        | amounts               | contracts              | amounts                | contracts              | amounts              | contracts              | amounts             | contracts            |
| <i>Pregrowing-season RZSM deficits</i> |                       |                        |                        |                        |                      |                        |                     |                      |
| All Legal Amazon municipalities        | 0.4249<br>(0.2971)    | 0.0019<br>(0.0820)     | 0.1074<br>(0.0970)     | 0.0183<br>(0.0883)     | 0.1521<br>(0.2745)   | -0.2268**<br>(0.1002)  | -0.2661<br>(0.1895) | -0.0397<br>(0.0297)  |
| Municipalities with more family farms  | 0.6141***<br>(0.1676) | -0.0780<br>(0.0742)    | -0.1933**<br>(0.0810)  | -0.2639***<br>(0.0566) | 1.0152**<br>(0.4974) | 0.0875<br>(0.0779)     | 0.0141<br>(0.0353)  | 0.0006<br>(0.0028)   |
| Municipalities with less family farms  | 0.3607<br>(0.3775)    | 0.0260<br>(0.1016)     | 0.1981**<br>(0.0989)   | 0.1064<br>(0.0788)     | -0.1176<br>(0.1666)  | -0.3247***<br>(0.1053) | -0.3531<br>(0.2347) | -0.0522<br>(0.0365)  |
| Difference                             | 0.2534<br>(0.4065)    | -0.1040<br>(0.1251)    | -0.3914***<br>(0.1211) | -0.3703***<br>(0.0968) | 1.1328**<br>(0.5232) | 0.4122***<br>(0.1302)  | 0.3672<br>(0.2392)  | 0.0528<br>(0.0370)   |
| <i>Growing-season RZSM deficits</i>    |                       |                        |                        |                        |                      |                        |                     |                      |
| All Legal Amazon municipalities        | -0.0131**<br>(0.0063) | -0.0052***<br>(0.0015) | -0.0152**<br>(0.0071)  | -0.0056***<br>(0.0015) | 0.0008<br>(0.0053)   | -0.0021<br>(0.0015)    | 0.0002<br>(0.0014)  | -0.00001<br>(0.0001) |
| Municipalities with more family farms  | -0.0200**<br>(0.0077) | -0.0078***<br>(0.0022) | -0.0232**<br>(0.0102)  | -0.0094***<br>(0.0018) | -0.0050<br>(0.0053)  | -0.0038*<br>(0.0020)   | -0.0008<br>(0.0011) | -0.0001<br>(0.0001)  |
| Municipalities with less family farms  | -0.0007<br>(0.0105)   | -0.0007<br>(0.0015)    | -0.0011<br>(0.0040)    | 0.0012<br>(0.0014)     | 0.0111<br>(0.0100)   | 0.0010<br>(0.0015)     | 0.0019<br>(0.0034)  | 0.0001<br>(0.0003)   |
| Difference                             | -0.0193<br>(0.0126)   | -0.0071***<br>(0.0026) | -0.0221**<br>(0.0107)  | -0.0106***<br>(0.0022) | -0.0161<br>(0.0111)  | -0.0049**<br>(0.0024)  | -0.0028<br>(0.0034) | -0.0002<br>(0.0003)  |
| lagged outcome                         | YES                   | YES                    | YES                    | YES                    | YES                  | YES                    | YES                 | YES                  |
| controls                               | YES                   | YES                    | YES                    | YES                    | YES                  | YES                    | YES                 | YES                  |
| fixed effects                          | YES                   | YES                    | YES                    | YES                    | YES                  | YES                    | YES                 | YES                  |
| N                                      | 8,283                 | 8,283                  | 8,283                  | 8,283                  | 3,845                | 3,845                  | 3,845               | 3,845                |

Note: <sup>a</sup>We use the same regressions as Table C8 except that we replace year-round RZSM deficits with pregrowing- and growing-season RZSM deficits. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## D — Main Figures

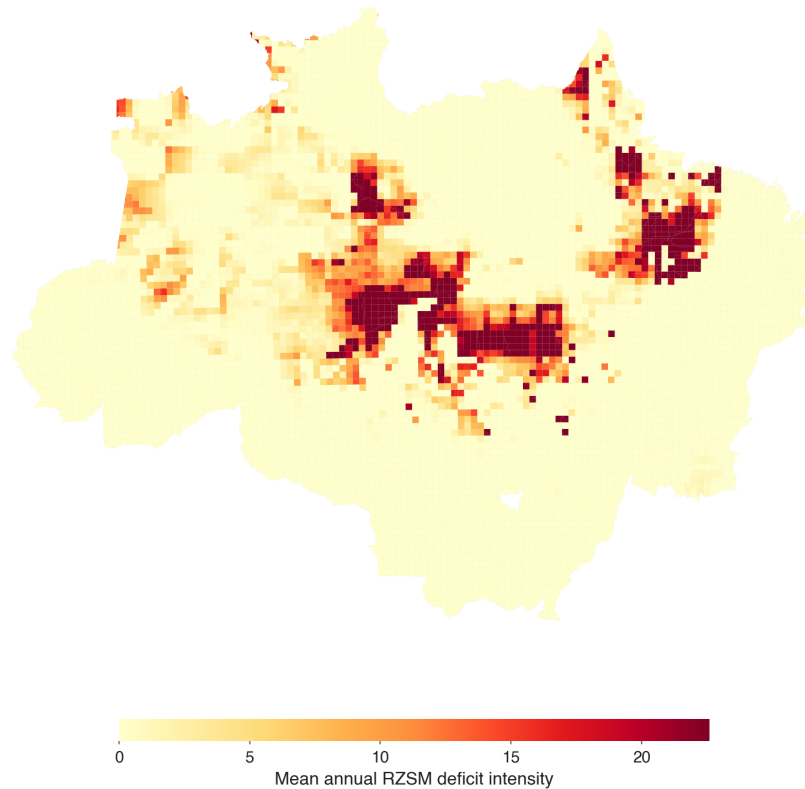
Where family farms dominate the frontier



Split at the median 2006-Census family-farm area share (~30%). RZSM analysis grid (~0.25°, ~25 km pixels), Brazilian Legal Amazon, 2005–2019.

**Figure D1 — High- and Low-Family-Farm Municipalities across the Legal Amazon.** Municipalities are split at the 2006 Agricultural Census median share of agricultural area in family-farm establishments. High-family-farm municipalities in red have family-farm area shares above the median while low-family-farm municipalities in blue have family-farm area shares below the median.

Average drought intensity (RZSM deficit)



The treatment. RZSM analysis grid ( $\sim 0.25^\circ$ ,  $\sim 25$  km pixels), Brazilian Legal Amazon, 2005–2019.

**Figure D2 — Mean Annual Drought Intensity, 2005–2019.** Grid-level cumulative deficits of standardized daily root-zone soil moisture relative to its long-run bottom-decile threshold. For each grid, we calculate annual deficits by summing the positive differences between the latter and the former across days within a year. Darker grids experienced more severe and/or more prolonged soil-moisture stress over the study period.

## E — Additional Tables

**Table E1 — Impacts of RZSM Deficits on Deforestation and Forest-related Environmental Fines at the Extensive Margin<sup>a</sup>**

|                                          | deforestation        |                       | environmental fines    |                        |
|------------------------------------------|----------------------|-----------------------|------------------------|------------------------|
|                                          | avg. effects         | hetero. effects       | avg. effects           | hetero. effects        |
| all Legal Amazon municipalities          | 0.00004<br>(0.00005) |                       | 0.00128**<br>(0.00051) |                        |
| municipalities with more family farms    |                      | 0.00006*<br>(0.00003) |                        | 0.00151**<br>(0.00071) |
| municipalities with less family farms    |                      | 0.00002<br>(0.00010)  |                        | 0.00090*<br>(0.00051)  |
| difference between the two               |                      | 0.00004<br>(0.00011)  |                        | 0.00060<br>(0.00084)   |
| one lag of the outcome variable          | YES                  | YES                   | YES                    | YES                    |
| one lag of forest stock ( <i>asinh</i> ) | YES                  | YES                   | YES                    | YES                    |
| price controls                           | YES                  | YES                   | YES                    | YES                    |
| policy controls                          | YES                  | YES                   | YES                    | YES                    |
| year fixed effects                       | YES                  | YES                   | YES                    | YES                    |
| municipality-by-RZSM-grid fixed effects  | YES                  | YES                   |                        |                        |
| municipality fixed effects               |                      |                       | YES                    | YES                    |
| <i>adj.R</i> <sup>2</sup>                | 0.82                 | 0.82                  | 0.45                   | 0.45                   |
| within <i>R</i> <sup>2</sup>             | 0.07                 | 0.07                  | 0.01                   | 0.01                   |
| <i>N</i>                                 | 184,240              | 184,030               | 10,794                 | 10,780                 |

*Note:* <sup>a</sup>Same regressions as in Table B1 and B2 except that we replace deforestation and fine outcomes with their dummies. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Table E2 — Impacts of Seasonal RZSM Deficits on Deforestation and Forest-Related Environmental Fines<sup>a</sup>**

|                                        | deforestation |                 | fines amount |                 | fines count  |                 |
|----------------------------------------|---------------|-----------------|--------------|-----------------|--------------|-----------------|
|                                        | avg. effects  | hetero. effects | avg. effects | hetero. effects | avg. effects | hetero. effects |
| <i>Pregrowing-season RZSM deficits</i> |               |                 |              |                 |              |                 |
| all Legal Amazon                       | -0.00079**    |                 | 0.1249       |                 | 0.0557       |                 |
| municipalities                         | (0.00035)     |                 | (0.5315)     |                 | (0.1005)     |                 |
| municipalities with                    |               | -0.00058***     |              | -1.4477***      |              | -0.1868***      |
| more family farms                      |               | (0.00020)       |              | (0.2085)        |              | (0.0372)        |
| municipalities with                    |               | -0.00124*       |              | 0.6420          |              | 0.1353          |
| less family farms                      |               | (0.00068)       |              | (0.4532)        |              | (0.0967)        |
| difference between the two             |               | 0.00066         |              | -2.0897***      |              | -0.3221***      |
|                                        |               | (0.00068)       |              | (0.4989)        |              | (0.1037)        |
| <i>Growing-season RZSM deficits</i>    |               |                 |              |                 |              |                 |
| all Legal Amazon                       | 0.00014*      |                 | 0.0184***    |                 | 0.0034***    |                 |
| municipalities                         | (0.00008)     |                 | (0.0067)     |                 | (0.0012)     |                 |
| municipalities with                    |               | 0.00028***      |              | 0.0230***       |              | 0.0046***       |
| more family farms                      |               | (0.00009)       |              | (0.0088)        |              | (0.0016)        |
| municipalities with                    |               | -0.00005        |              | 0.0117          |              | 0.0016          |
| less family farms                      |               | (0.00009)       |              | (0.0077)        |              | (0.0014)        |
| difference between the two             |               | 0.00032***      |              | 0.0113          |              | 0.0030          |
|                                        |               | (0.00012)       |              | (0.0113)        |              | (0.0020)        |
| lagged outcome                         | YES           | YES             | YES          | YES             | YES          | YES             |
| controls                               | YES           | YES             | YES          | YES             | YES          | YES             |
| fixed effects                          | YES           | YES             | YES          | YES             | YES          | YES             |
| <i>adj.R</i> <sup>2</sup>              | 0.89          | 0.89            | 0.56         | 0.56            | 0.69         | 0.69            |
| within <i>R</i> <sup>2</sup>           | 0.29          | 0.29            | 0.01         | 0.01            | 0.03         | 0.03            |
| <i>N</i>                               | 184,240       | 184,030         | 10,794       | 10,780          | 10,794       | 10,780          |

*Note:* <sup>a</sup>We use the same deforestation and environmental fine measures as in Table B1 and B2. But we replace year-round RZSM deficits with two seasonal RZSM deficits, one for the pregrowing season (July-August) and one for the growing season (all other months in the same year). Regressions include municipality and year fixed effects, lagged dependent variable, and various controls as the other tables above. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .



## F — Robustness Results

**Table F1 — The Association Between Local Deforestation and Local Moisture Inflow<sup>a</sup>**

|                               | coefficient           | clustered standard errors |                 |                 |
|-------------------------------|-----------------------|---------------------------|-----------------|-----------------|
|                               | estimates             | moisture grid             | Conley (200 km) | Conley (500 km) |
| local deforestation           | -0.0041               | (0.0040)                  | (0.0042)        | (0.0058)        |
| one lag of local forest stock | 0.1398 <sup>*b</sup>  | (0.0820)                  | (0.0688)        | (0.0826)        |
| one lag of local moisture     | 0.1086 <sup>***</sup> | (0.0149)                  | (0.0231)        | (0.0404)        |
| year fixed effects            | YES                   | YES                       | YES             | YES             |
| grid fixed effects            | YES                   | YES                       | YES             | YES             |
| <i>adj.R</i> <sup>2</sup>     | 0.92                  | 0.92                      | 0.92            | 0.92            |
| within <i>R</i> <sup>2</sup>  | 0.01                  | 0.01                      | 0.01            | 0.01            |
| <i>N</i>                      | 6,305                 | 6,305                     | 6,305           | 6,305           |

*Note:* <sup>a</sup>We obtain moisture flow data from Zhang et al. (2025). Local moisture refers to moisture that originates from and is absorbed within the same grid cell. <sup>b</sup>We indicate statistical significance level based on the standard errors clustered at the moisture grid level, which are listed in the parentheses next to the estimated coefficients. We also report Conley spatial clustered standard errors at 200-km and 500-km distance. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Table F2 — The Association Between Local Deforestation and Moisture Inflow at Different Spatial Degrees**

| degree | coefficient | clustered standard errors |                 |                 |                  |
|--------|-------------|---------------------------|-----------------|-----------------|------------------|
|        |             | moisture grid             | Conley (200 km) | Conley (500 km) | Conley (1000 km) |
| 0      | -0.0041     | (0.0040)                  | (0.0042)        | (0.0058)        | (0.0069)         |
| 1      | 0.0004      | (0.0046)                  | (0.0049)        | (0.0068)        | (0.0081)         |
| 3      | -0.0019     | (0.0045)                  | (0.0047)        | (0.0064)        | (0.0076)         |
| 5      | -0.0050     | (0.0044)                  | (0.0046)        | (0.0063)        | (0.0072)         |
| 7      | -0.0063     | (0.0044)                  | (0.0045)        | (0.0063)        | (0.0072)         |
| 9      | -0.0058     | (0.0043)                  | (0.0045)        | (0.0063)        | (0.0072)         |

*Note:* The dependent variable is log moisture inflow at the grid level (1° resolution). Independent variable of interest is forest loss relative to the base year of 2000 (transformed by *asinh* as before). All regressions include grid and year fixed effects, lagged dependent variable, and lagged forest stock. Standard errors are clustered at the grid level (first column) or corrected for spatial correlation using a Bartlett kernel with the indicated distance cutoff (km). Number of observations: 6,305; number of grid clusters: 485. None of the coefficients are statistically significant at the 10% level.

**Table F3 — The Association Between Local Deforestation and Rainfall<sup>a</sup>**

|                               | coefficient | clustered standard errors |                 |                 |
|-------------------------------|-------------|---------------------------|-----------------|-----------------|
|                               | estimates   | municipality              | Conley (200 km) | Conley (500 km) |
| local deforestation           | -0.0001     | (0.0023)                  | (0.0032)        | (0.0045)        |
| one lag of local forest stock | 0.0093      | (0.0119)                  | (0.0219)        | (0.0415)        |
| one lag of local rainfall     | 0.0606***   | (0.0220)                  | (0.0246)        | (0.0305)        |
| year fixed effects            | YES         | YES                       | YES             | YES             |
| grid fixed effects            | YES         | YES                       | YES             | YES             |
| <i>adj.R</i> <sup>2</sup>     | 0.80        | 0.92                      | 0.92            | 0.92            |
| within <i>R</i> <sup>2</sup>  | 0.05        | 0.01                      | 0.01            | 0.01            |
| <i>N</i>                      | 167,146     | 167,146                   | 167,146         | 167,146         |

*Note:* <sup>a</sup>We obtain precipitation data from TerraClimate and aggregate them at our municipality-RZSM grid level. There are 11,939 grids in 771 Legal Amazon municipalities that are matched. <sup>b</sup>We indicate statistical significance level based on the standard errors clustered at the municipality level, which are listed in the parentheses next to the estimated coefficients. We also report Conley spatial clustered standard errors at 200-km and 500-km distance. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Table F4 — Conley Standard Errors for The Drought-Induced Deforestation Results**

|                    | coefficient | clustered standard errors |                 |                 |
|--------------------|-------------|---------------------------|-----------------|-----------------|
|                    | estimates   | municipality              | Conley (200 km) | Conley (500 km) |
| all municipalities | 0.00014*    | (0.00008)                 | (0.00008)       | (0.00009)       |
| more family farms  | 0.00028***  | (0.00009)                 | (0.00010)       | (0.00010)       |
| less family farms  | -0.00005    | (0.00009)                 | (0.00010)       | (0.00010)       |

*Note:* We report three standard errors in parentheses. One is clustered at the municipality level and two are Conley spatial HAC standard errors with 200 km and 500 km cutoffs. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Table F5 — Impacts of Year-Round RZSM Deficits on Deforestation and Forest-Related Environmental Fines with Continuous Family-Farm Area Shares<sup>a</sup>**

|                                                                     | deforestation          | fines amount       | fines count          |
|---------------------------------------------------------------------|------------------------|--------------------|----------------------|
| Municipalities with 0% family-farm area<br>(fully commercial farms) | -0.00001<br>(0.00011)  | 0.0085<br>(0.0092) | 0.00158<br>(0.00165) |
| Municipalities with 100% family-farm area<br>(fully family farms)   | 0.00045**<br>(0.00020) | 0.0367<br>(0.0245) | 0.00685<br>(0.00444) |
| Marginal effect of family-farm share<br>(0% → 100%)                 | 0.00046*<br>(0.00027)  | 0.0282<br>(0.0313) | 0.00527<br>(0.00568) |
| lagged outcome                                                      | YES                    | YES                | YES                  |
| controls                                                            | YES                    | YES                | YES                  |
| fixed effects                                                       | YES                    | YES                | YES                  |
| <i>adj.R</i> <sup>2</sup>                                           | 0.89                   | 0.56               | 0.69                 |
| within <i>R</i> <sup>2</sup>                                        | 0.29                   | 0.01               | 0.03                 |
| <i>N</i>                                                            | 184,030                | 10,780             | 10,780               |

*Note:* <sup>a</sup>We use the same deforestation and environmental fine measures as in Table B1 and B2. Family-farm exposure is measured continuously as the municipality’s family-farm area share (0–1), interacted with year-round RZSM deficit intensity. Rows report the estimated effect at 0% and 100% family-farm share (via `lincom`), and the marginal effect of moving across the full range. Regressions include municipality and year fixed effects, lagged dependent variable, and various controls as the other tables above. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Table F6 — Impacts of Year-Round RZSM Deficits on Deforestation: Family-Farm and Rural-Poor-Household Heterogeneity<sup>a</sup>**

|                                          | (1) family farm         | (2) poor household     | (3) interaction         |
|------------------------------------------|-------------------------|------------------------|-------------------------|
| Municipalities with less family farms    | -0.00005<br>(0.00009)   |                        |                         |
| Municipalities with more family farms    | 0.00027***<br>(0.00009) |                        |                         |
| Municipalities with less poor households |                         | 0.00004<br>(0.00010)   |                         |
| Municipalities with more poor households |                         | 0.00022**<br>(0.00010) |                         |
| Less family farms, less poor households  |                         |                        | -0.00006<br>(0.00012)   |
| More family farms, less poor households  |                         |                        | 0.00023**<br>(0.00011)  |
| Less family farms, more poor households  |                         |                        | -0.00001<br>(0.00007)   |
| More family farms, more poor households  |                         |                        | 0.00029***<br>(0.00011) |
| lagged outcome                           | YES                     | YES                    | YES                     |
| controls                                 | YES                     | YES                    | YES                     |
| fixed effects                            | YES                     | YES                    | YES                     |
| <i>adj.R</i> <sup>2</sup>                | 0.89                    | 0.89                   | 0.89                    |
| within <i>R</i> <sup>2</sup>             | 0.30                    | 0.30                   | 0.30                    |
| <i>N</i> <sup>b</sup>                    | 180,726                 | 180,726                | 180,726                 |

*Note:* <sup>a</sup>We obtain the rural household income distribution data (Tabela 1993) in the 2000 Demographic Census from the Brazilian Institute of Geography and Statistics. The dataset records municipality-level household counts by monthly income bins relative to the minimum monthly wage rate. We classify a rural household with its monthly income less than the minimum wage as a poor household, we define municipalities with rural-poor-household shares above the median, roughly 40%, as those with more rural poor households. Column (1) interacts year-round RZSM deficit intensity with the more-family-farm municipality indicator only; Column (2) interacts it with the more-poor-household municipality indicator only; Column (3) includes both indicators and their triple interaction with year-round RZSM deficit intensity. Reported coefficients are linear combinations giving the estimated effect for each named subgroup. Regressions include municipality and year fixed effects, lagged dependent variable, and various controls as the other tables above. <sup>b</sup>We are only able to match the rural household income distribution data with 755 Legal Amazon municipalities in our main sample. So, the number of observations slightly drops. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Table F7 — Impacts of Year-Round RZSM Deficits on Deforestation: Family-Farm and Public-Land Heterogeneity<sup>a</sup>**

|                                       | (1) family farm         | (2) public land       | (3) interaction         |
|---------------------------------------|-------------------------|-----------------------|-------------------------|
| Municipalities with less family farms | -0.00005<br>(0.00009)   |                       |                         |
| Municipalities with more family farms | 0.00028***<br>(0.00009) |                       |                         |
| Municipalities with less public land  |                         | 0.00004<br>(0.00023)  |                         |
| Municipalities with more public land  |                         | 0.00014*<br>(0.00008) |                         |
| Less family farms, less public land   |                         |                       | -0.00026<br>(0.00019)   |
| More family farms, less public land   |                         |                       | 0.00090<br>(0.00063)    |
| Less family farms, more public land   |                         |                       | -0.00002<br>(0.00010)   |
| More family farms, more public land   |                         |                       | 0.00026***<br>(0.00009) |
| lagged outcome                        | YES                     | YES                   | YES                     |
| controls                              | YES                     | YES                   | YES                     |
| fixed effects                         | YES                     | YES                   | YES                     |
| $adj.R^2$                             | 0.89                    | 0.89                  | 0.89                    |
| within $R^2$                          | 0.29                    | 0.29                  | 0.29                    |
| $N$                                   | 184,030                 | 184,030               | 184,030                 |

*Note:* <sup>a</sup>We obtain the public land data from the CNFP (Cadastro Nacional de Florestas Públicas, 2024 edition, rectified July 2025). The dataset includes public land (mostly forest) that is either designated or undesignated (not yet allocated to a specific use). We focus on undesignated public land that is less likely to be protected. We classify municipalities undesignated public land area shares above the median as municipalities with more public land. Column (1) interacts year-round RZSM deficit intensity with the more-family-farm municipality indicator only; Column (2) interacts it with the more-public-land municipality indicator only; Column (3) includes both indicators and their triple interaction with year-round RZSM deficit intensity. Reported coefficients are linear combinations giving the estimated effect for each named subgroup. Regressions include municipality and year fixed effects, lagged dependent variable, and various controls as the other tables above. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

**Table F8 — Impacts of Year-Round Monthly-SPEI-Based Drought Deficits on Deforestation and Forest-Related Environmental Fines<sup>a</sup>**

|                                       | deforestation         |                       | fines amount       |                     | fines count        |                    |
|---------------------------------------|-----------------------|-----------------------|--------------------|---------------------|--------------------|--------------------|
|                                       | avg. effects          | hetero. effects       | avg. effects       | hetero. effects     | avg. effects       | hetero. effects    |
| all Legal Amazon municipalities       | 0.0162***<br>(0.0037) |                       | 0.0147<br>(0.2319) |                     | 0.0265<br>(0.0483) |                    |
| municipalities with more family farms |                       | 0.0152***<br>(0.0036) |                    | -0.0554<br>(0.2613) |                    | 0.0330<br>(0.0546) |
| municipalities with less family farms |                       | 0.0190**<br>(0.0087)  |                    | 0.2031<br>(0.4168)  |                    | 0.0132<br>(0.0838) |
| difference between the two            |                       | -0.0038<br>(0.0088)   |                    | -0.2585<br>(0.4642) |                    | 0.0198<br>(0.0932) |
| lagged outcome                        | YES                   | YES                   | YES                | YES                 | YES                | YES                |
| controls                              | YES                   | YES                   | YES                | YES                 | YES                | YES                |
| fixed effects                         | YES                   | YES                   | YES                | YES                 | YES                | YES                |
| <i>adj.R</i> <sup>2</sup>             | 0.89                  | 0.89                  | 0.56               | 0.56                | 0.71               | 0.71               |
| within <i>R</i> <sup>2</sup>          | 0.29                  | 0.29                  | 0.01               | 0.01                | 0.03               | 0.03               |
| <i>N</i> <sup>b</sup>                 | 182,546               | 182,336               | 10,654             | 10,640              | 10,654             | 10,640             |

*Note:* <sup>a</sup>We use the same deforestation and environmental fine measures as in Table B1 and B2, but replace daily RZSM-based drought deficits with year-round monthly SPEI-based drought deficits. To align with the resolution of our RZSM-based drought measure, we obtain the 0.25-degree SPEI data from Liu et al. (2024). <sup>b</sup>We obtain monthly SPEI data for 761 municipalities, so the number of observations drops. As before, one municipality is missing in the Agricultural Census data when it comes to heterogeneous effects. Regressions include municipality and year fixed effects, lagged dependent variable, and various controls as the other tables above. Standard errors, clustered at the municipality level, are in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

# G — Carbon Valuation of Drought-Induced Deforestation

This appendix translates the estimated effect of drought on deforestation (Section 6, Table B1) into a monetary valuation of the associated carbon-emission externality. The goal is to express the marginal social damage of drought-induced forest loss in dollar terms, providing a policy-relevant benchmark against which the cost of the rural finance interventions discussed in the main text can be compared. The valuation proceeds in three steps: (i) recovering the physical forest loss in hectares implied by the estimated asinh-scale response; (ii) converting hectares cleared into committed gross carbon-dioxide emissions; and (iii) applying a range of social-cost-of-carbon (SCC) values to monetize those emissions. We treat the resulting figures as a lower bound on total social damage, since they omit belowground and soil carbon, co-emitted greenhouse gases from associated burning, and non-carbon ecosystem-service losses.

## G.1. From the asinh response to hectares cleared

Our deforestation outcome is the inverse hyperbolic sine of absolute annual forest loss,  $Y_{i,t} = \text{asinh}(h_{i,t})$ , where  $h_{i,t}$  denotes hectares of forest loss in municipality-RZSM grid cell  $i$  in year  $t$ . Because the outcome is nonlinear in hectares, the estimated coefficient  $\beta$  does not map directly into a level effect; we therefore recover predicted drought-attributable hectares from the fitted model rather than relying on a local approximation.

Let  $\hat{Y}_{i,t}$  denote the fitted value from the main specification evaluated at the observed drought intensity  $DI_{i,t}$ , and let  $\hat{Y}_{i,t}^0$  denote the corresponding counterfactual fitted value obtained by setting the drought term to a no-drought reference level  $DI^0$  (e.g.,  $DI^0 = 0$ , or the cell-specific historical median) while holding all other covariates, the lagged outcome (2005 only), and the fixed effects at their observed values. Back-transforming out of asinh units, predicted forest loss with and without the drought shock is

$$\hat{h}_{i,t} = \sinh(\hat{Y}_{i,t}), \quad \hat{h}_{i,t}^0 = \sinh(\hat{Y}_{i,t}^0), \quad (\text{G.1})$$

and the drought-attributable forest loss in cell  $i$  and year  $t$  is the difference

$$\Delta h_{i,t} = \hat{h}_{i,t} - \hat{h}_{i,t}^0 = \sinh(\hat{Y}_{i,t}) - \sinh(\hat{Y}_{i,t}^0). \quad (\text{G.2})$$

Aggregating over the cells and years of interest yields the total drought-attributable hectares cleared,

$$\Delta H = \sum_{i \in \mathcal{S}} \sum_t \Delta h_{i,t}, \quad (\text{G.3})$$

where  $\mathcal{S}$  is the set of cells over which the valuation is conducted—for example, all cells, the high-family-farm subsample, or the cells exposed to a specific drought episode such as 2005, 2010, or 2015–2016.

## G.2. From hectares to committed carbon-dioxide emissions

We convert cleared area into committed gross CO<sub>2</sub> emissions following the standard biomass-accounting approach. Aboveground biomass (AGB) is converted to carbon using the IPCC default carbon fraction  $\phi = 0.47$  for tropical forest (IPCC 2019), and carbon is converted to carbon dioxide using the molecular-weight ratio  $44/12 \approx 3.67$ . Treating all aboveground carbon on cleared land as committed emissions, the per-hectare emission factor is

$$EF = \text{AGB} \times \phi \times \frac{44}{12} \quad (\text{t CO}_2 \text{ ha}^{-1}). \quad (\text{G.4})$$

For the Brazilian Amazon we adopt the LiDAR-calibrated basin-average biomass of  $\text{AGB} = 174 \text{ t ha}^{-1}$  from Ometto et al. (2023), equivalent to roughly  $82 \text{ t C ha}^{-1}$  and an emission factor of about  $300 \text{ t CO}_2 \text{ ha}^{-1}$ . This is a deliberately conservative central value: it counts only aboveground live biomass and excludes belowground biomass, deadwood, litter, and soil organic carbon, which can add substantially to total committed emissions where clearing is followed by burning or where organic soils are involved. Total committed emissions from drought-attributable clearing are then

$$E = \Delta H \times EF. \quad (\text{G.5})$$

Table G1 reports  $EF$  under low, central, and high carbon-density assumptions spanning the range reported for Amazonian forest.

**Table G1 — Committed gross CO<sub>2</sub> emission factor per hectare of forest cleared**

| Carbon-density scenario                     | AGB (t/ha) | Carbon (t C/ha) | $EF$ (t CO <sub>2</sub> /ha) |
|---------------------------------------------|------------|-----------------|------------------------------|
| Low (degraded / transitional)               | 110        | 52              | 190                          |
| Central (basin average, Ometto et al. 2023) | 174        | 82              | 300                          |
| High (dense primary forest)                 | 300        | 141             | 517                          |

*Notes: Carbon fraction  $\phi = 0.47$  (IPCC 2019); CO<sub>2</sub>/C ratio  $44/12 \approx 3.67$ . Figures rounded. Estimates count aboveground live biomass only and therefore understate total committed emissions.*

## G.3. Monetizing emissions with the social cost of carbon

The dollar value of the carbon externality from drought-induced clearing is the product of committed emissions and the social cost of carbon,

$$D = E \times \text{SCC} = \Delta H \times EF \times \text{SCC}. \quad (\text{G.6})$$

There is no single agreed SCC, and the choice materially affects the valuation, so we report a range that brackets the values most used in policy and academic work. Our central, policy-relevant benchmark is the U.S. Environmental Protection Agency’s official schedule (U.S. Environmental Protection Agency 2023), which uses a modular framework with dynamic Ramsey discounting calibrated to three near-term rates. For a 2020 emissions pulse, the



EPA central SC-CO<sub>2</sub> is \$190 per metric ton at the 2.0% near-term discount rate, with the discount-rate range spanning \$120 (2.5%) to \$340 (1.5%). For comparison we report two lower integrated-assessment-model benchmarks—the U.S. interagency working group interim central estimate of about \$51 per ton (Interagency Working Group on Social Cost of Greenhouse Gases 2021), and Nordhaus’s DICE estimates (\$31 per ton for 2015 emissions in DICE-2016R, Nordhaus 2017; \$50 per ton on the cost–benefit-optimal path and \$66 baseline for 2020 in DICE-2023, Barrage and Nordhaus 2024)—and Pindyck’s survey-based estimates as an independent, non-IAM benchmark, which yield an average SCC of roughly \$80–\$100 per ton after trimming outliers and \$200 per ton or higher for the untrimmed expert distribution (Pindyck 2019). These benchmarks are summarized in Table G2.

**Table G2 — Social-cost-of-carbon benchmarks (2020 dollars per metric ton CO<sub>2</sub>)**

| Source                                                                          | SCC (\$/t CO <sub>2</sub> ) | Basis                                               |
|---------------------------------------------------------------------------------|-----------------------------|-----------------------------------------------------|
| Nordhaus (Nordhaus 2017), DICE-2016R                                            | 31                          | 2015 emissions (2010 US\$); baseline path           |
| IWG (Interagency Working Group on Social Cost of Greenhouse Gases 2021) interim | 51                          | 2020 emissions; 3% constant discount rate           |
| Barrage and Nordhaus (Barrage and Nordhaus 2024), DICE-2023                     | 50–66                       | 2020 emissions; \$50 optimal, \$66 baseline         |
| Pindyck (Pindyck 2019), trimmed                                                 | 80–100                      | Expert survey with outliers trimmed                 |
| EPA (U.S. Environmental Protection Agency 2023) central                         | 190                         | 2020 emissions; 2.0% near-term Ramsey discount rate |
| EPA (U.S. Environmental Protection Agency 2023), 1.5% rate                      | 340                         | 2020 emissions; 1.5% near-term Ramsey discount rate |
| Pindyck (Pindyck 2019), untrimmed                                               | ≥200                        | Full expert survey distribution                     |

*Notes: Estimates use different base years and discount rates and are not inflation-harmonized. Values should be converted to a common base year in the final draft. IAM = integrated assessment model.*

## G.4. Illustrative valuation and sensitivity

To make the magnitudes concrete, Table G3 applies Equation (G.6) to the illustrative placeholder of  $\Delta H = 10,000 \text{ ha yr}^{-1}$  of drought-attributable forest loss (to be replaced with the sample-based estimate from Equation (G.3)). At the central carbon density of  $300 \text{ t CO}_2 \text{ ha}^{-1}$ , 10,000 ha implies 3.0 Mt CO<sub>2</sub> of committed emissions, valued at about \$570 million at the EPA central SCC of \$190, and ranging from roughly \$93 million at Nordhaus’s \$31 figure to about \$1.0 billion at the EPA \$340 value. Because the valuation is linear in  $\Delta H$ , the entire table rescales by the single factor  $(\Delta H/10,000)$  once the sample-based hectare estimate is inserted.

The spread across the table shows that the carbon-density assumption and the SCC choice are the dominant sources of uncertainty—together they move the valuation by roughly an order of magnitude—whereas the estimated deforestation response enters only through  $\Delta H$ .

**Table G3 — Annual carbon-damage value of 10,000 ha of drought-induced clearing (US\$ millions)**

| Emission factor                     | SCC (\$/t CO <sub>2</sub> ) |      |       |       |
|-------------------------------------|-----------------------------|------|-------|-------|
|                                     | \$31                        | \$51 | \$190 | \$340 |
| Low (190 t CO <sub>2</sub> /ha)     | 59                          | 97   | 361   | 646   |
| Central (300 t CO <sub>2</sub> /ha) | 93                          | 153  | 570   | 1,020 |
| High (517 t CO <sub>2</sub> /ha)    | 160                         | 264  | 982   | 1,758 |

*Notes: Cell entries are  $\Delta H \times EF \times SCC$  in millions of US\$, for  $\Delta H = 10,000 \text{ ha yr}^{-1}$  (illustrative placeholder). Damages scale linearly in  $\Delta H$ ; rescale by  $(\Delta H/10,000)$  for the sample-based estimate.*

We therefore report the central combination alongside the full range rather than a single point estimate.

## G.5. Interpretation and link to policy

The valuation should be read as the marginal external carbon cost of the deforestation that drought induces, concentrated—as the main results show—among family-farm municipalities. Because this clearing appears to function as a short-run coping response to drought-induced income shocks rather than as forward-looking commercial land conversion, the carbon damages estimated here are, in principle, avoidable through interventions that relax the underlying liquidity constraint. The relevant policy comparison is therefore between the per-ton carbon damage computed above and the per-ton cost of providing the rural insurance or credit access that would blunt the coping motive. If the cost of such programs per ton of avoided CO<sub>2</sub> lies below the SCC range in Table G2, they are justified on carbon grounds alone, before accounting for their direct welfare benefits to vulnerable smallholders.

Three caveats apply. First, the asinh-to-hectares conversion in Equation (G.2) relies on the estimated model being correctly specified off the observed drought support; we recommend reporting  $\Delta H$  for more than one counterfactual  $DI^0$  as a robustness check. Second, the SCC benchmarks use different base years and discount rates and should be harmonized to a common base year. Third, the committed-emissions accounting assumes full release of aboveground carbon; depending on the post-clearing land use documented in the “Land Use after Drought” analysis, a lag or decay adjustment may be warranted.